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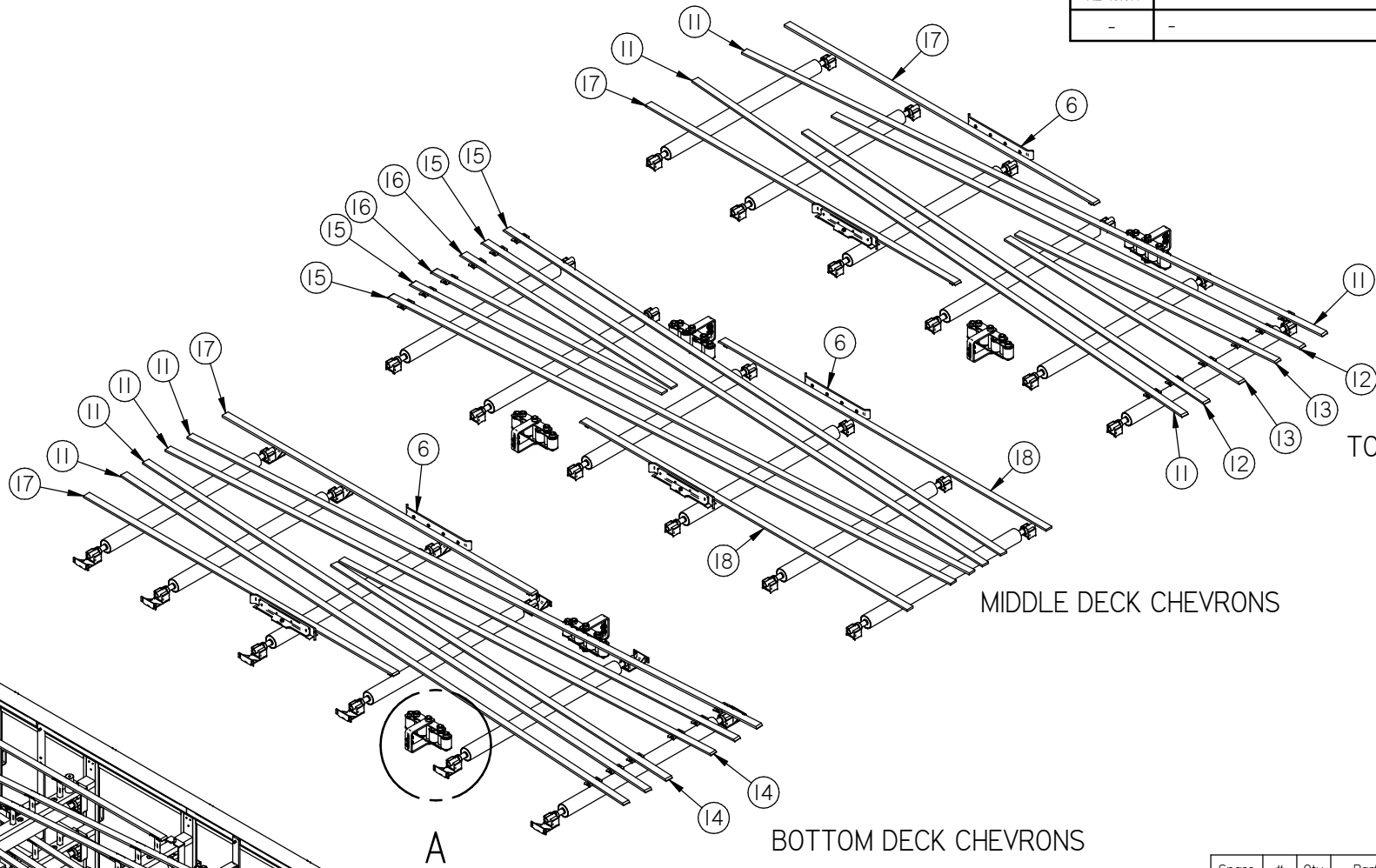
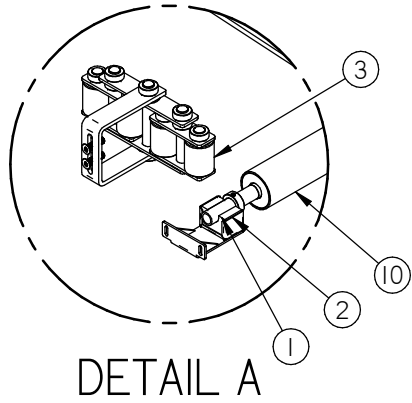
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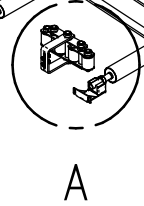
REVISION	CHANGE	DATE
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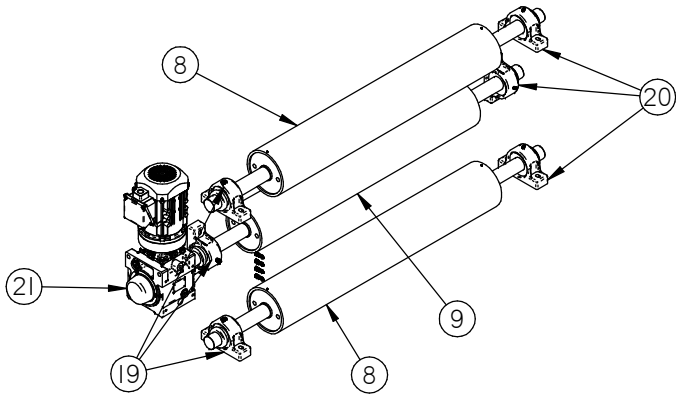
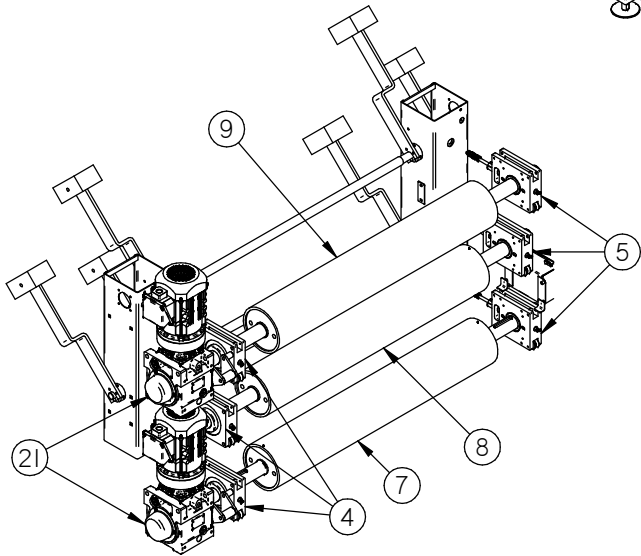
TOP DECK CHEVRONS

MIDDLE DECK CHEVRONS

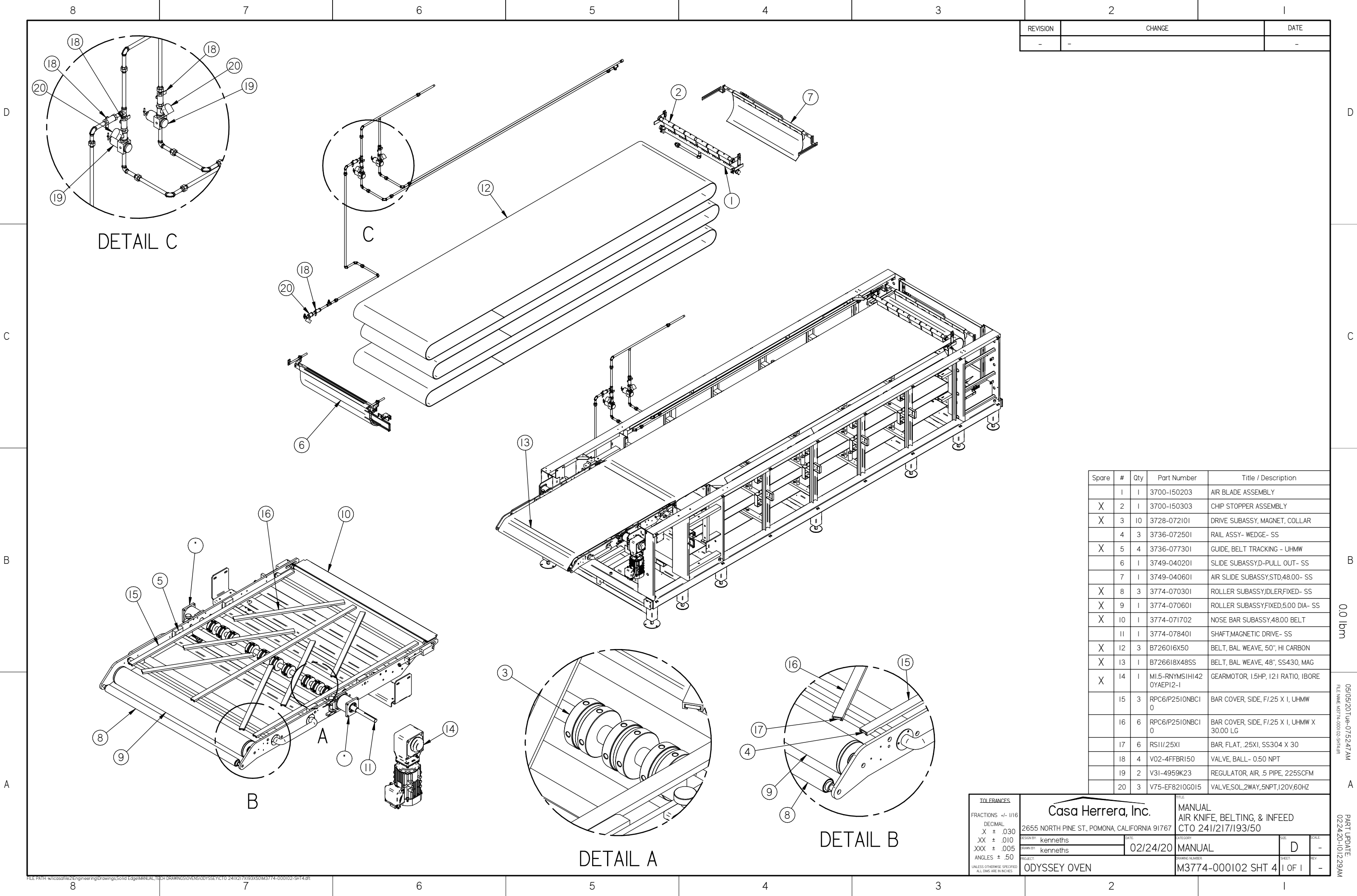
BOTTOM DECK CHEVRONS



Spare	#	Qty	Part Number	Title / Description
X	1	35	3713-021701	BEARING, INSERT, GRAPHITE
	2	35	3736-120903	HOLDER SUBASSY,GRAPH MT- SS
X	3	24	3736-122601	ROLLER SUBASSY- CS (HARDENED)
X	4	3	3736-170101	BEARING SUBASSY,TAKE UP
X	5	3	3736-170102	BEARING SUBASSY,TAKE UP,EXP
X	6	6	3741-120601	GUIDE SUBASSY,ANGLE- CS
X	7	1	3749-020801	ROLLER SUBASSY,DRIVE- 8.50 X 50.00
X	8	3	3749-020901	ROLLER SUBASSY,DRIVEN-
X	9	2	3749-021001	ROLLER ASSY,DRIVE- 8.50
X	10	18	3749-120701	ROLLER SUBASSY,BELT SUPP
	11	6	3774-121701	V-GUIDE ASSY,CHEV, MT,182.00-SS
	12	2	3774-121702	V-GUIDE ASSY,CHEV, MT,148.00-SS
	13	2	3774-121703	V-GUIDE ASSY,CHEV, MT,82.00-SS
	14	2	3774-121704	V-GUIDE ASSY,CHEV, MT,120.00-SS
	15	4	3774-121705	V-GUIDE ASSY,CHEV, MT,180.00-SS
	16	2	3774-121706	V-GUIDE ASSY,CHEV, MT,75.00-SS
	17	4	3774-122201	CHEVRON SUBASSY,BELT CARRY- 105.00
	18	2	3774-122203	CHEVRON SUBASSY,BELT CARRY- 111.00
X	19	3	BPB22439H	BEARING,PILLOW BLOCK- 2.438 BORE
X	20	3	BPEB22439H	BEARING,PILLOW BLOCK,EXP- 2.438 BORE
X	21	3	M3-LHYMS33AI20 EPY228-2.18	GEARMOTOR, 3HP, 281, 2.18 BORE



TOLERANCES		Casa Herrera, Inc.		TITLE	
FRACTIONS +/- 1/16	DECIMAL	2655 NORTH PINE ST, POMONA, CALIFORNIA 91767		MANUAL	
X ± .030	XX ± .010	DESIGN BY kenneths	DATE 02/24/20	CATEGORY	SIZE
XXX ± .005	ANGLES ± .50	DRAWN BY kenneths	PROJECT	MANUAL	D
UNLESS OTHERWISE SPECIFIED ALL DIMS ARE IN INCHES		ODYSSEY OVEN		SHEET	
		M3774-000102 SHT 1		REV	
		1 OF 1		-	



REVISION	CHANGE	DATE
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Spare	#	Qty	Part Number	Title / Description
	1	1	3700-150203	AIR BLADE ASSEMBLY
X	2	1	3700-150303	CHIP STOPPER ASSEMBLY
X	3	10	3728-072101	DRIVE SUBASSY, MAGNET, COLLAR
	4	3	3736-072501	RAIL ASSY- WEDGE- SS
X	5	4	3736-077301	GUIDE, BELT TRACKING - UHMW
	6	1	3749-040201	SLIDE SUBASSY,D-PULL OUT- SS
	7	1	3749-040601	AIR SLIDE SUBASSY,STD,48.00- SS
X	8	3	3774-070301	ROLLER SUBASSY,DLR/FIXED- SS
X	9	1	3774-070601	ROLLER SUBASSY,FIXED,5.00 DIA- SS
X	10	1	3774-071702	NOSE BAR SUBASSY,48.00 BELT
	11	1	3774-078401	SHAFT,MAGNETIC DRIVE- SS
X	12	3	B726016X50	BELT, BAL WEAVE, 50", HI CARBON
X	13	1	B726618X48SS	BELT, BAL WEAVE, 48", SS430, MAG
X	14	1	M1.5-RNYMSIHI42 OYAEP12-1	GEARMOTOR, 1.5HP, 12:1 RATIO, IBORE
	15	3	RPC6/P2510NBCI 0	BAR COVER, SIDE, F/25 X 1, UHMW
	16	6	RPC6/P2510NBCI 0	BAR COVER, SIDE, F/25 X 1, UHMW X 30.00 LG
	17	6	RS11/25X1	BAR, FLAT, .25X1, SS304 X 30
	18	4	V02-4FFBR150	VALVE, BALL- 0.50 NPT
	19	2	V31-4959K23	REGULATOR, AIR, 5 PIPE, 225SCFM
	20	3	V75-EF8210G015	VALVE,SOL,2WAY,5NPT,120V,60HZ

TOLERANCES		FRACTIONS +/- 1/16		DECIMAL		ANGLES ± .50		UNLESS OTHERWISE SPECIFIED ALL DIMS ARE IN INCHES	
X ± .030		XX ± .010		XXX ± .005					
DESIGN BY		kenneths		DATE		02/24/20		CATEGORY	
DRAWN BY		kenneths		PROJECT		ODYSSEY OVEN		DRAWING NUMBER	
M3774-000102 SHT 4		SHEET		1 OF 1		REV			

Casa Herrera, Inc.		2655 NORTH PINE ST., POMONA, CALIFORNIA 91767		TITLE		MANUAL		AIR KNIFE, BELTING, & INFEEED	
						CTO 241/217/193/50			

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REVISION	CHANGE	DATE
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Spare	#	Qty	Part Number	Title / Description
	1	1	3728-038501	ROD,RETAINER, 0.50 DIA- SS
	2	4	3728-112101	INSULATION SUBASSY,
	3	1	3728-112701	PANEL SUBASSY, REMOVEABLE
	4	1	3728-112706	Insulation Panel Assy,No Handle
	5	2	3728-114801	BRACKET,INSUL. HOLDER MT- SS
	6	1	3728-165502	BRKT,INSULATION. MT, COMB- SS
	7	1	3728-165602	BRKT,INSULATION MT, COMB- SS
	8	1	3729-111001	PART. SUBASSY- REAR,TOP- SS,RT
	9	1	3729-111006	PART. SUBASSY- REAR,TOP- SS,LT
	10	1	3729-111101	PART SUBASSY-REAR,BOT-SS,RT
	11	1	3729-111106	PART. SUBASSY, REAR,BOT- SS,LT
	12	1	3729-111401	PART. SUBASSY, REAR,MID- SS,RT
	13	1	3729-111406	PART. SUBASSY, REAR,MID- SS,LT
	14	1	3729-111506	PART. SUB,SWING DOOR- SS,RT
	15	1	3729-111601	PARTITION SUBASSY,FIXED- REAR,
	16	1	3729-112001	PARTITION SUBASSY, FT,TOP-
	17	1	3729-112006	PARTITION SUBASSY, FT,TOP-
	18	2	3737-111201	PARTITION SUBASSY,FIXED,REAR,BOT-SS
	19	2	3737-111301	PARTITION SUBASSY,REAR,MID,RT- SS
	20	1	3741-111308	PARTITION SUBASSY,FIXED- REAR,
	21	2	3744-112601	INSULATION SUBASSY,
	22	1	3744-112606	INSULATION SUBASSY,
	23	1	3749-051801	Rear Door Assy, Pull Off
	24	1	3749-051802	Rear Door Assy, Pull Off
	25	4	3749-112610	COVER,BURNER CUTOUTS,6 ROW- SS
	26	1	3749-112612	COVER,BURNER CUTOUTS,6 ROW- SS
	27	1	3749-112901	INSULATION SUBASSY,FIRE WALL
	28	2	3750-111706	PARTITION SUBASSY- FT,BOT- SS
	29	1	3750-111801	PARTITION SUBASSY-FT,MID-SS,LT
	30	1	3750-111806	PARTITION SUBASSY- FT,MID- SS,
	31	2	3750-111901	PARTITION SUBASSY,FIXED- FT,
	32	2	3754-111701	PARTITION SUBASSY,BOT,REAR- SS
	33	1	3755-112501	PARTITION SUBASSY,BOT- SS
	34	1	3772-112701	PANEL SUBASSY, REMOVEABLE
	35	2	SP-2710	SPRING, .6090D X 3.5 LG, .513 ID,

TOLERANCES		FRACTIONS +/- 1/16		DECIMAL		XX ± .030		XXX ± .005		ANGLES ± .50		UNLESS OTHERWISE SPECIFIED		ALL DIMS ARE IN INCHES	
DESIGN BY		kenneths		DATE		02/24/20		CATEGORY		MANUAL		SHEET		1 OF 1	
DRAWN BY		kenneths		PROJECT		ODYSSEY OVEN		DRAWING NUMBER		M3774-000102 SHT 5		REV		-	

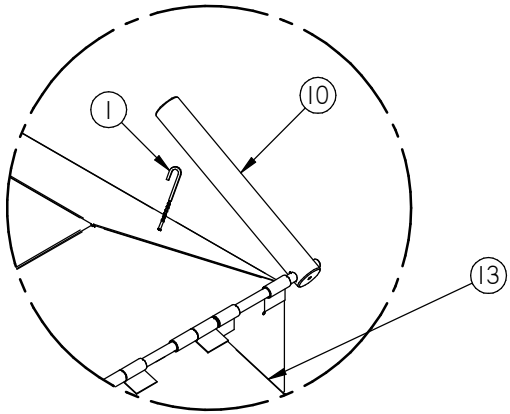
Casa Herrera, Inc.

2655 NORTH PINE ST, POMONA, CALIFORNIA 91767

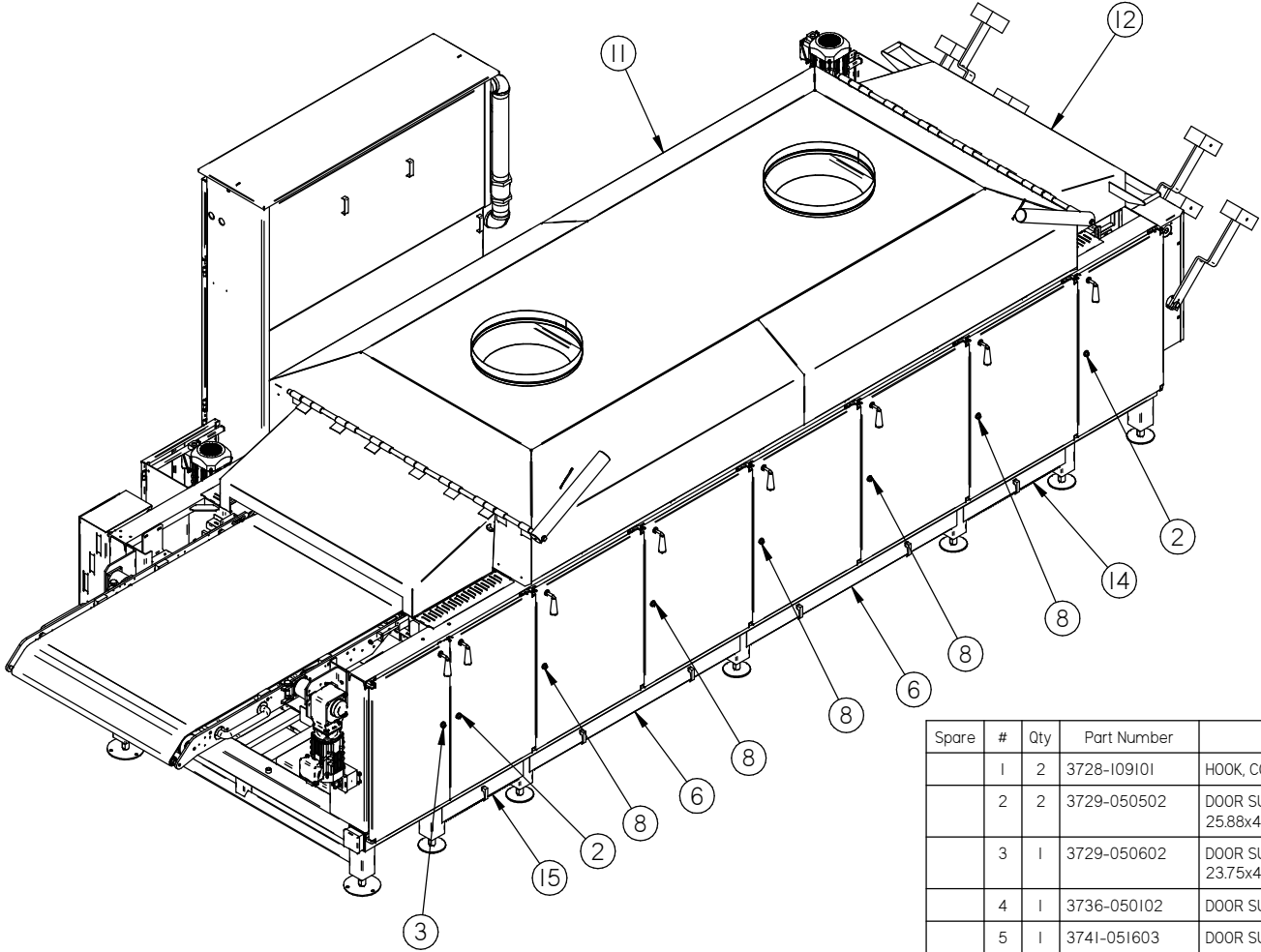
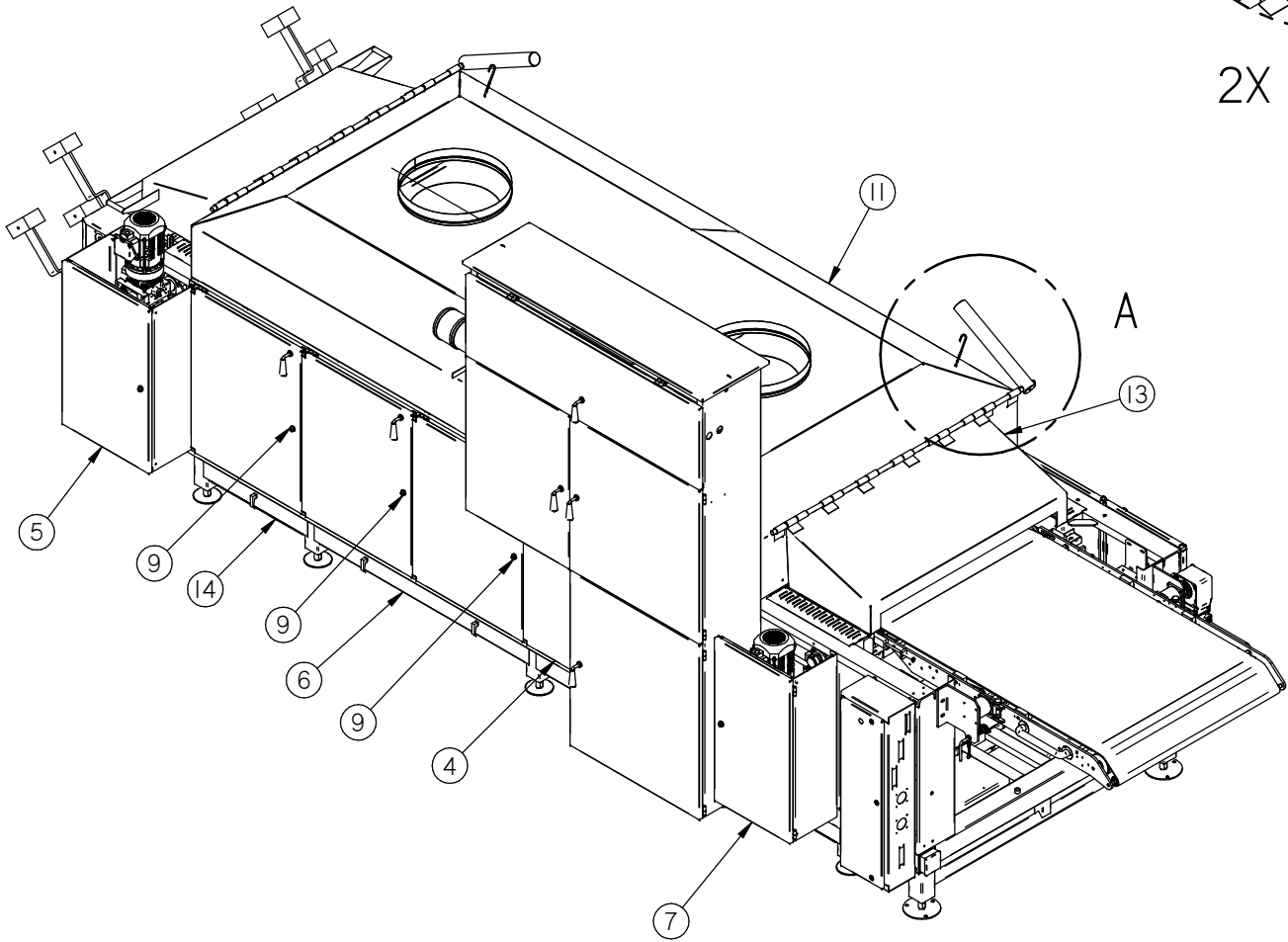
MANUAL
INSULATION
CTO 241/217/193/50

PART UPDATE:
02/24/20-10:54:43AM

REVISION	CHANGE	DATE
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2X DETAIL A



Spare	#	Qty	Part Number	Title / Description
	1	2	3728-109101	HOOK, COUNTER WEIGHT
	2	2	3729-050502	DOOR SUBASSY, BEARING COMPT- 25.88x42- SS
	3	1	3729-050602	DOOR SUBASSY, FRAME EXT- 23.75x42.00- SS
	4	1	3736-050102	DOOR SUBASSY, INSULATED, SM
	5	1	3741-051603	DOOR SUBASSY, MTR COMPT
	6	4	3747-130501	DOOR SUBASSY, CRUMB TRAY, 64W-SS
	7	1	3750-051601	DOOR SUBASSY, MOTOR COMPT- 25.75x42.00
	8	5	3752-050101	DOOR SUBASSY, INSUL, CE- SS
	9	3	3752-050102	DOOR SUBASSY, INSUL, CE- SS
	10	2	3772-108302	COUNTER WEIGHT, HOOD END, 24.00 LG
	11	1	3774-100102	HOOD SUBASSY- SS
	12	1	3774-100701	EXTENSION SUBASSY, DISCH END- SS
	13	1	3774-100803	EXTENSION SUBASSY, INFEED END- SS
	14	2	3774-131001	DOOR SUBASSY, SM- SS
	15	2	3774-131002	DOOR SUBASSY, CRUMB, SM- SS

TOLERANCES

FRACTIONS +/- 1/16

DECIMAL

X ± .030

XX ± .010

XXX ± .005

ANGLES ± .50

UNLESS OTHERWISE SPECIFIED

ALL DIMS ARE IN INCHES

Casa Herrera, Inc.

2655 NORTH PINE ST., POMONA, CALIFORNIA 91767

DESIGN BY: kenneths

DRAWN BY: kenneths

PROJECT: ODYSSEY OVEN

DATE: 02/24/20

CATEGORY: MANUAL

DRAWING NUMBER: M3774-000102 SHT 6

TITLE: MANUAL CRUMB TRAYS, DOORS, & HOOD CTO 241/217/193/50

SIZE: D

SCALE: -

SHEET: 1 OF 1

REV: -

ARTESANO FOODS, SO# 090031.002 ELECTRICAL SYSTEM, OVEN, 460V/60HZ						
M2M #92056900C						
Electrical Bill of Materials						
	PART NUMBER	DESCRIPTION	QUANTITY	U/M	MANUFACTURER	
	3728-221003	ENCLOSURE SUBASSY, PUSH BUTTON	3	EA	CASA HERRERA	
	E001-OR120IB0000105	MODULE, ORDAN IGNITION, 120 VAC	1	EA	ORDAN	
	E002-3569	NIPPLE, INSULATING, SCANNER, W/LOGO	1	EA	FIRE EYE	
	E002-871TMBX21N3	SENSOR, PROXIMITY, 20-132VAC/DC, 2	3	EA	ALLEN-BRADLEY	
	E002-889NF3AFC6F	CORDSET, QUICK DISCONNECT, MINI	3	EA	ALLEN-BRADLEY	
	E002-LFC4HU1000	CONTROL, TEMP, 120V, TYPE J, W/SPAD	2	EA	WATLOW	
	E002-T40LT30SF00	SENSOR HEAD, 2WIRE, 4-20MA	3	EA	FLUKE	
	E002-TXSLTSF	TX SENSOR HEAD	3	EA	RAYTEK	
	E002-UV1A6	SCANNER, FLAME CONTROL, 560-610VAC	1	EA	FIRE EYE	
	E003-3872K81	THERMOCOUPLE, TYPE J, .1875 DIA"	2	EA	MCMASTER-CARR	
	E004-194EE3217536N	SWITCH ASSY, DISCONNECT, 32A, 480VA	4	EA	ALLEN-BRADLEY	
	E004-HBLDS3AC	SWITCH, DISCONNECT, 30A, 3 POLE	1	EA	HUBBELL	
	E005-800FALP	LATCH, PLASTIC	1	EA	ALLEN-BRADLEY	
	E005-800FD5C	MODULE, INCANDENCENT, LATCH MT	1	EA	ALLEN-BRADLEY	
	E005-800FPLF0	BUTTON, PUSH, MOMENTARY, ILLUMINATE	1	EA	ALLEN-BRADLEY	
	E005-800FX01	BLOCK, CONTACT, N.C., LATCH MT, SCR	1	EA	ALLEN-BRADLEY	
	E005-800TFXTQH2RA5	BUTTON, PUSH-PULL/TWIST, RED	4	EA	ALLEN-BRADLEY	
	E005-800TX646	PLATE, NAME, "EMERGENCY STOP"	4	EA	ALLEN-BRADLEY	
	E005-800TXD1	BLOCK, CONTACT, SHALLOW, 1NO	4	EA	ALLEN-BRADLEY	
	E006-48996	TRANSFORMER, IGNITION, 120V, 60HZ	1	EA	DUNGS	
	E007-3240K12	KIT, CONNECTOR END, PIN&SKT, 10POLE	1	EA	HARTING	
	E007-3240K14	KIT, CONNECTOR END, PIN&SKT, 24POLE	4	EA	MCMASTER-CARR	
	E007-3240K15	KIT, CONNECTOR END, PIN&SKT, 6POLE	1	EA	HARTING	
	E007-A404NFSS	ENCLOSURE, J BOX, TYPE 4X, CLAMP	2	EA	HOFFMAN	
	E007-A8064CHNFSS	ENCLOSURE, J BOX, TYPE 4X, HINGED	2	EA	HOFFMAN	
	E007-A8P6	PANEL, SUB, J BOX, 6.75X4.88, PAINT	1	EA	HOFFMAN	
	E008-09330242601	CONNECTOR, INSERT MODULE, MALE	1	EA	HARTING	
	E008-09330242701	CONNECTOR, INSERT MODULE, FEMALE	2	EA	HARTING	
	E008-1411319	BASE, MT, PANEL, SIZE B6, GRAY	2	EA	MCMASTER-CARR	
	E008-1604840	SLEEVE HOUSING, 24PIN	1	EA	PHOENIX CONTACT	
	E008-1645493	MOUNTING BASE, B24	5	EA	PHOENIX CONTACT	
	E008-3030213	PLUG, IN BRIDGE, TEN POSN, 24AMP	5	EA	PHOENIX CONTACT	
	E008-3030417	COVER, END FOR ST2.5 TERMINAL BL	10	EA	PHOENIX CONTACT	
	E008-3031212	TERMINAL, BLOCK, SPRING CAGE, GRAY	93	EA	PHOENIX CONTACT	
	E008-3031238	TERMINAL, BLOCK, GND MOD, GRN/YL	4	EA	PHOENIX CONTACT	
	E008-3031487	TERMINAL, BLOCK, SPRING CAGE, GRAY	18	EA	PHOENIX CONTACT	
	E008-3031500	TERMINAL, BLOCK, SPRING CAGE,	6	EA	PHOENIX CONTACT	
	E008-420R7W	RECEPTACLE, WATERTIGHT, 20A, 3 POLE	1	EA	HUBBELL	
	E008-430R7W	RECEPTACLE, WATERTIGHT, 30A, 3 POLE	1	EA	HUBBELL	
	E008-C2WH6	COVER, WIRING DUCT, 2", WHITE	2	FT	PANDUIT	
	E008-G2X3WH6	WIRING DUCT, PLSTC	2	FT	PANDUIT	
	E008-J20FFB	WIRE, THERMOCOUPLE, TYPE J, 20GA, W	200	FT	MCMASTER-CARR	
	E008-XBACPT6	BARRIER, END, TERM BLK, .32W X	15	EA	CUTTLE HAMMER	
	E34-M7284A1004	ACTUATOR, MODUTROL IV MOTOR, 120VAC	3	EA	HONEYWELL	
	E53-AAP	COLLAR, AIR PURGE	3	EA	FLUKE	

ARTESANO FOODS, SO# 090031.005 ELECTRICAL SYST., SH-CCFB-6 ROW, 460V/60HZ						
M2M #92056900						
Electrical Bill of Materials						
	PART NUMBER	DESCRIPTION	QUANTITY	U/M	MANUFACTURER	
	E001-1585JM8CBJM2	CABLE, ETHERNET, 8COND, STR, MALE	16	EA	ALLEN-BRADLEY	
	E001-1606XLE240EN	POWER SUPPLY, 24-28V DC, 240W	1	EA	ALLEN-BRADLEY	
	E001-1769ECR	CAP, TERMINATOR, RIGHT END	1	EA	ALLEN-BRADLEY	
	E001-1769IF8	MODULE, INPUT, 8 POINT	1	EA	ALLEN-BRADLEY	
	E001-1769IQ16	MODULE,INPUT,DIGITL DC,16IN,COMP LX	4	EA	ALLEN-BRADLEY	
	E001-1769L30ER	ETHERNET, DUAL, W/DLR CAP., 1MB	1	EA	ALLEN-BRADLEY	
	E001-1769OF4	MODULE, OUTPUT, 4 POINT	1	EA	ALLEN-BRADLEY	
	E001-1769OW16	MODULE,OUTPT,DIGTL,CNTC,16OT	1	EA	ALLEN-BRADLEY	
	E001-1769PA2	MODULE,PWRSPPL,120/220VAC,2A,COMPLX	1	EA	ALLEN-BRADLEY	
	E001-2711PT10C22D9P	TOUCH SCREEN, 10"SVGA, TFT, 24V DC	1	EA	ALLEN-BRADLEY	
	E001-2891933	SWITCH, ETHERNET, 16 PORT	1	EA	PHOENIX CONTACT	
	E001-5697T31	RECEPTACLE, PATH CORD, FEMALE, HVY	1	EA	MCMASTER CARR	
	E001-5697T44	COVER, JACK - DATA CONNECTOR	1	EA	MCMASTER CARR	
	E001-AEK115NDH	INTERLOCK, DOOR, 110/120V, 50/60HZ	1	EA	HOFFMAN	
	E001-EC514101PK	REMOTE ACCESS FOR PLC	1	EA	QUANTUM AUTOMATION	
	E001-KDRULA1L	LINE REACTOR, FOR: 2HP	2	EA	TCI	
	E001-KDRULA2L	LINE REACTOR, FOR: 3HP	8	EA	TCI	
	E001-KDRULA3L	LINE REACTOR, FOR: 5HP	1	EA	TCI	
	E001-KDRULA4L	LINE REACTOR, FOR: 7.5HP	1	EA	TCI	
	E001-MEP230	PROGRAMMER, F/MP5201	1	EA	FIRE EYE	
	E002-25BD010N104	INVERTER, DRIVE, 5HP, 480V AC INPUT	1	EA	ALLEN-BRADLEY	
	E002-25BD013N104	INVERTER, PWRFLEX525, 480VAC, 7.5HP	1	EA	ALLEN-BRADLEY	
	E002-25BD4P0N104	INVERTER, DRIVE, 2HP, 480V AC INPUT	2	EA	ALLEN-BRADLEY	
	E002-25BD6P0N104	INVERTER, AC DRIVE, 3HP, 380-480V	8	EA	ALLEN-BRADLEY	
	E002-MEC120R	CHASSIS, MICRO, W/RESET, 120VAC	1	EA	FIRE EYE	
	E002-MEUV4	AMPLIFIER, UV, 3 SEC FFRT	1	EA	FIRE EYE	
	E003-AEKDF	SWITCH, KEY, FOR DOOR INTERLOCK	1	EA	HOFFMAN	
	E004-100C16D10	CONTACTOR, 3 POLE, 120VAC	1	EA	ALLEN-BRADLEY	
	E004-100SC16D14C	CONTACTOR, SAFETY, 16A	1	EA	ALLEN-BRADLEY	
	E004-140GHRVM12B	HANDLE, ROTARY, 12" - BLACK	1	EA	ALLEN-BRADLEY	
	E004-140GJ3F3D15	BREAKER, CIRCUIT, 160A, FRAME, 3POL	1	EA	ALLEN-BRADLEY	
	E004-140GJMTL63	TERMINAL LUG,MULTI,FRAME J,6X14-2	1	EA	ALLEN-BRADLEY	
	E004-140GJTLC13	LUG, TERMINAL, FRAME-J, 1X14-1/0AWG	1	EA	ALLEN-BRADLEY	
	E004-140MC2EB25	PROTECTOR, MOTOR, MANUAL, 600V	6	EA	ALLEN-BRADLEY	
	E004-140MC2EB63	PROTECTOR, MOTOR, MANUAL, 600V	2	EA	ALLEN-BRADLEY	

	PART NUMBER	DESCRIPTION	QUANTITY	U/M	MANUFACTURER	
	E004-140MC2EC10	PROTECTOR, MOTOR, MANUAL, 600V	8	EA	ALLEN-BRADLEY	
	E004-140MC2EC16	PROTECTOR, MOTOR, MANUAL, 600V	1	EA	ALLEN-BRADLEY	
	E004-140MC2EC20	PROTECTOR, MOTOR, MANUAL, 600V	1	EA	ALLEN-BRADLEY	
	E004-140MCAFA10	CONTACT, AUXILIARY, FRT MT, 1 N.O.	18	EA	ALLEN-BRADLEY	
	E004-1489M1C020	BREAKER, CIRCUIT, 2A, 1 POLE, MINI	4	EA	ALLEN-BRADLEY	
	E004-1489M1C030	BREAKER, CIRCUIT, 3A, 1POLE, MINI	1	EA	ALLEN-BRADLEY	
	E004-1489M1C100	BREAKER, CIRCUIT, 10A, 1 POLE	1	EA	ALLEN-BRADLEY	
	E004-1489M1C250	BREAKER, CIRCUIT, 25A, 1 POLE	1	EA	ALLEN-BRADLEY	
	E004-1489M1D030	BREAKER, CIRCUIT, 3A, MINIATURE	1	EA	ALLEN-BRADLEY	
	E004-1489M2D080	BREAKER, CIRCUIT, 8A, 2POLE, 400V	1	EA	ALLEN-BRADLEY	
	E004-440RN23126	RELAY, SAFETY, INPUTS, 1NC/3NO	4	EA	ALLEN-BRADLEY	
	E004-4983DS480403	PROTECTOR,SURGE & FILTER,480V,40KA	1	EA	ALLEN-BRADLEY	
	E004-613060	BASE, FLAME CONTROL FIREYE	1	EA	FIRE EYE	
	E004-700HLT1U1	RELAY, TERMINAL BLOCK, 115/125V	13	EA	ALLEN-BRADLEY	
	E005-800TFXQH2GD2	BUTTON, PUSH/PULL, 2POS, 30MM	1	EA	ALLEN-BRADLEY	
	E005-800TX559	PLATE, NAME, LEGEND BLANK	1	EA	ALLEN-BRADLEY	
	E006-1492REC15	RECEPTACLE, DIN, RAIL, STD DUPLEX	1	EA	ALLEN-BRADLEY	
	E006-CE2000F008	TRANSFORMER, 2000VA, ENCAPSULATED	1	EA	ACME	
	E007-A72P72	PANEL, SUB, LG, NEMA 12, 68X68	1	EA	HOFFMAN	
	E007-A74H7212SSLP	ENCLOSURE, FLOOR MTG, 2 DOOR	1	EA	HOFFMAN	
	E007-GBK10	KIT, GROUND BAR, 10 CIRCUIT	4	EA	EATON	
	E008-199DR1	RAIL, DIN, 35MMX7.5MM, ZINC PLTD	2	EA	ALLEN-BRADLEY	
	E008-3030213	PLUG, IN BRIDGE, TEN POSN, 24AMP	10	EA	PHOENIX CONTACT	
	E008-3030417	COVER, END FOR ST2.5 TERMINAL BL	50	EA	PHOENIX CONTACT	
	E008-3031212	TERMINAL, BLOCK, SPRING CAGE, GRAY	113	EA	PHOENIX CONTACT	
	E008-3031238	TERMINAL, BLOCK, GND MOD, GRN/YL	3	EA	PHOENIX CONTACT	
	E008-3031487	TERMINAL, BLOCK, SPRING CAGE, GRAY	54	EA	PHOENIX CONTACT	
	E008-3031500	TERMINAL, BLOCK, SPRING CAGE,	13	EA	PHOENIX CONTACT	
	E008-800TN310	RING, PROT, F/2POS PUSH-PULL ILLUM	1	EA	ALLEN-BRADLEY	
	E008-C1.5WH6	COVER, WIRING DUCT, 1.5", WHITE	8	FT	PANDUIT	
	E008-C2WH6	COVER, WIRING DUCT, 2", WHITE	6	FT	PANDUIT	
	E008-G1.5X3WH6	WIRING DUCT, PAND	8	FT	PANDUIT	
	E008-G2X3WH6	WIRING DUCT, PLSTC	6	FT	PANDUIT	
	E008-XBACPT6	BARRIER, END, TERM BLK, .32W X	13	EA	CUTTLER HAMMER	
	E008-XBAES35C	ANCHOR, END, DIN RAIL, 1 SCREW MTD,	50	EA	CUTTLER HAMMER	
	V20-RZ08A1264X	AIR CONDITIONER, 120V, 8000 BTUH	1	EA	ISC	

ARTESANO FOODS, SO# 090031.006 ELECTRICAL SYSTEM, CABLE TRAY, 460V/60HZ

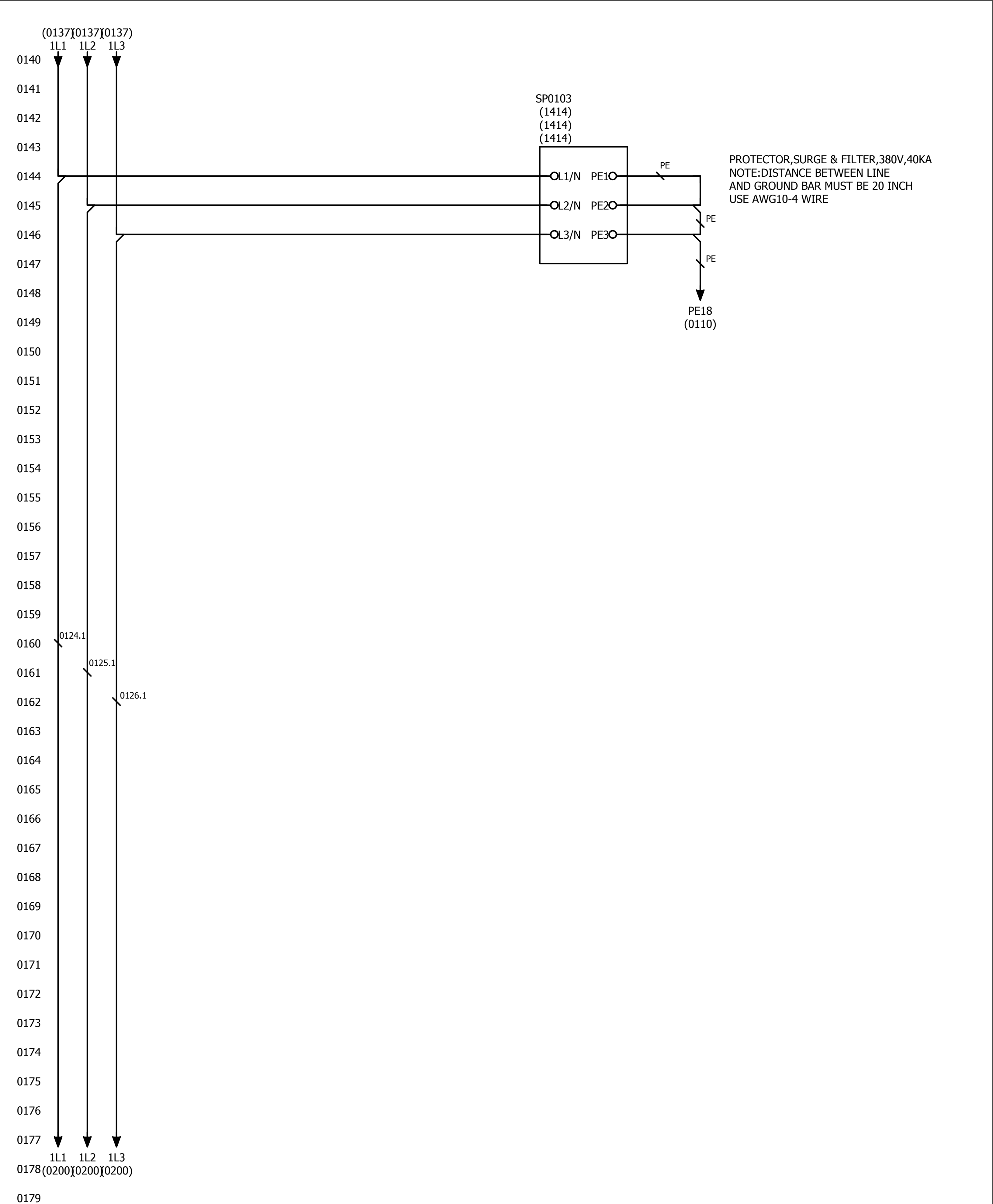
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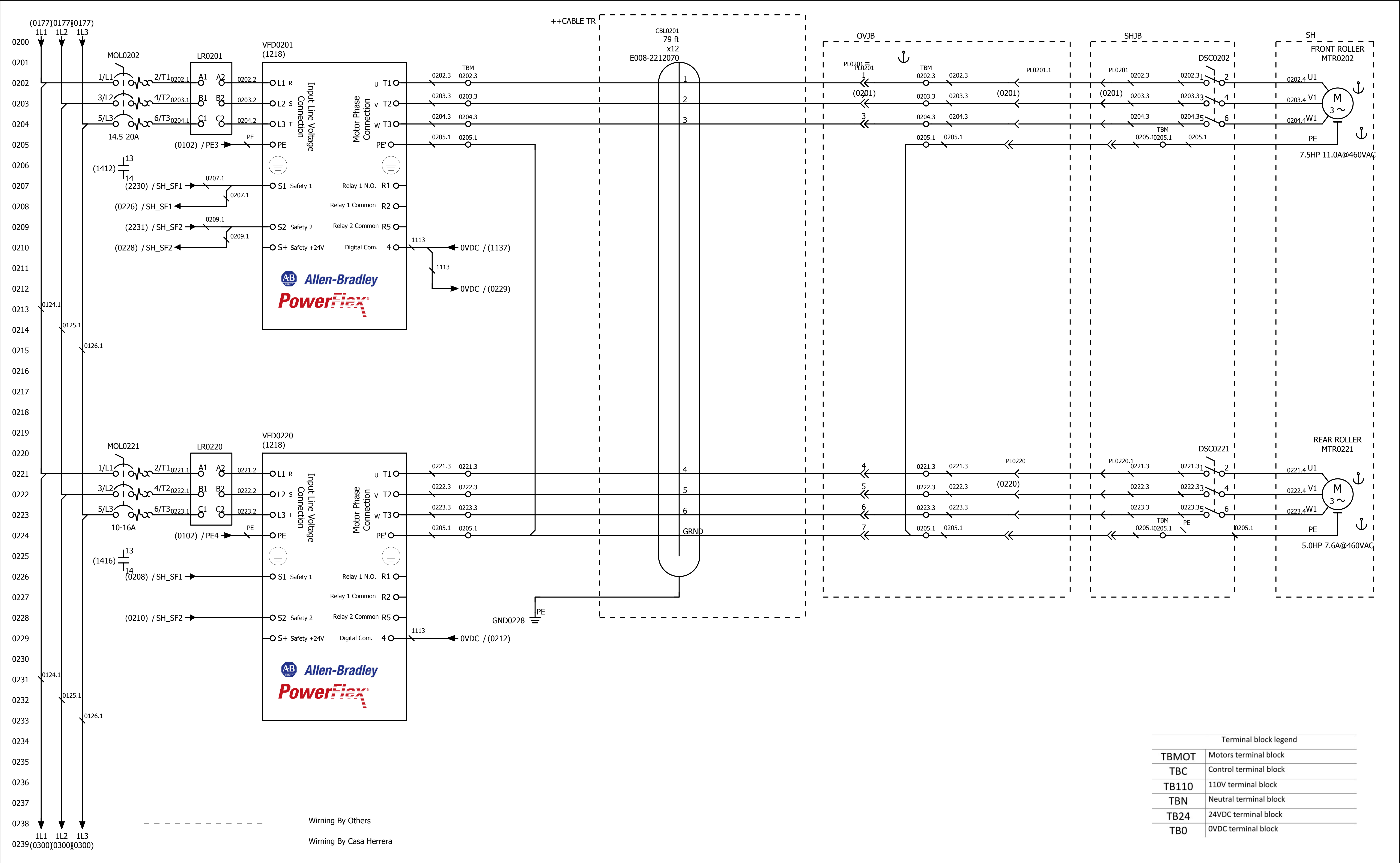
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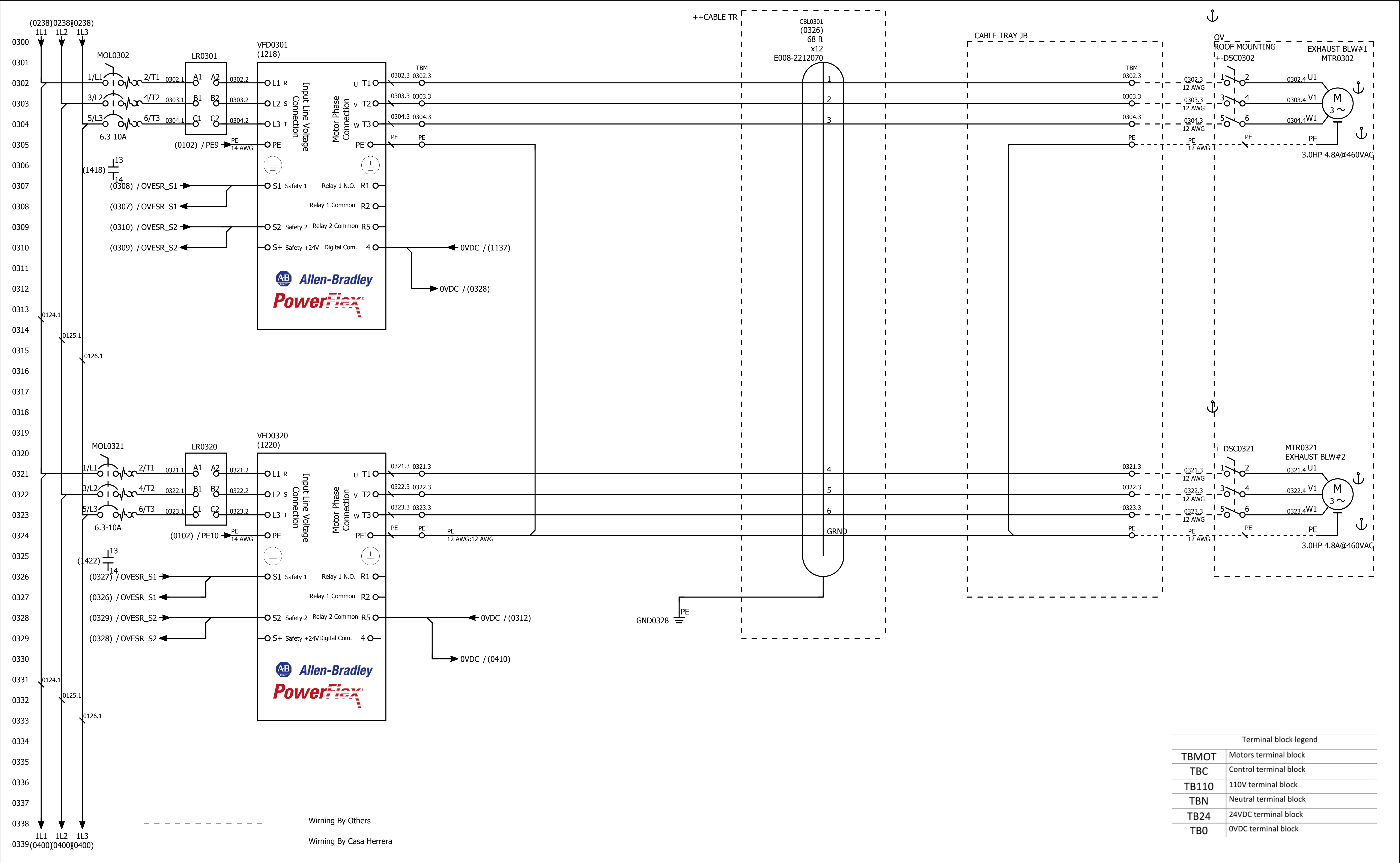
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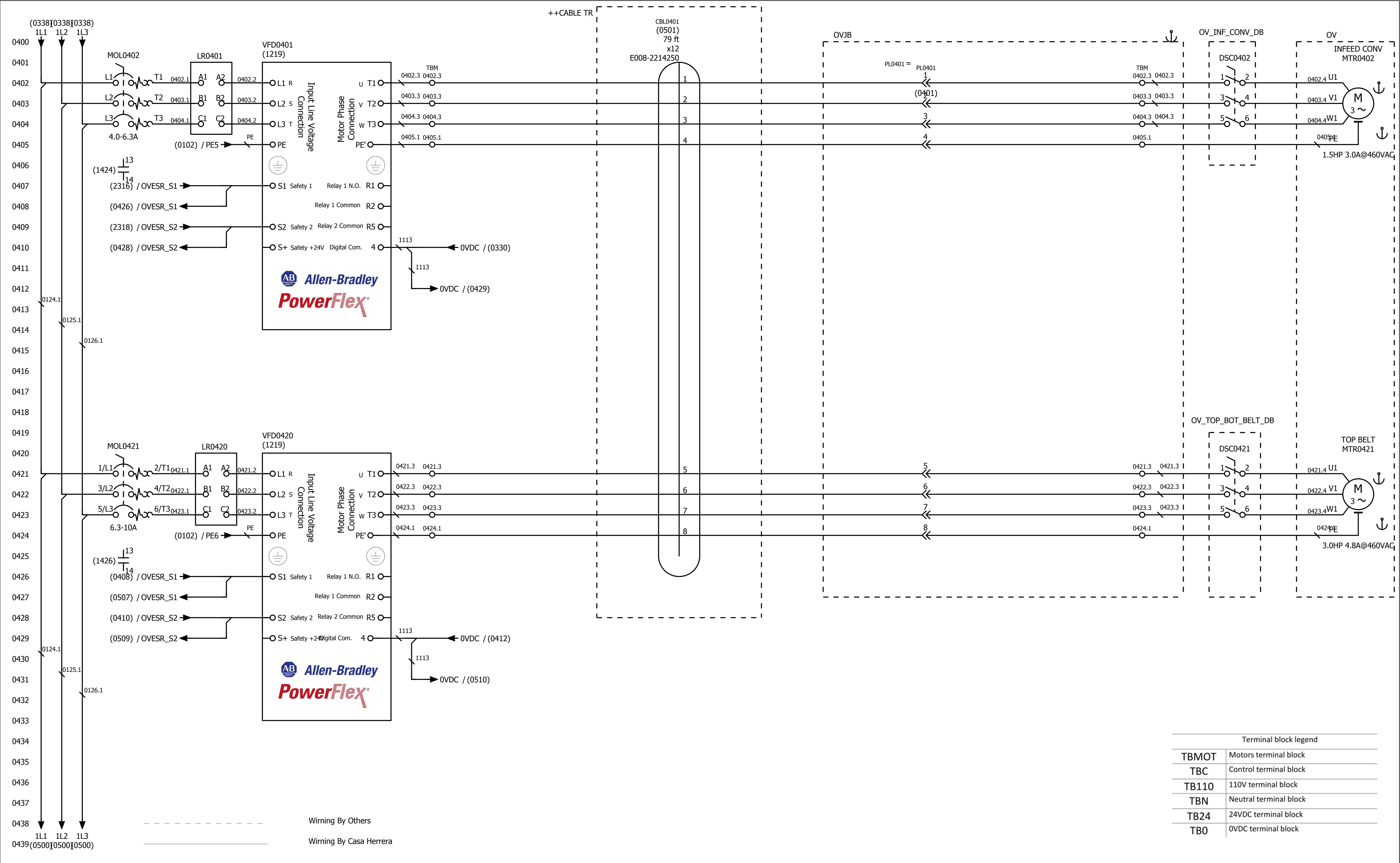
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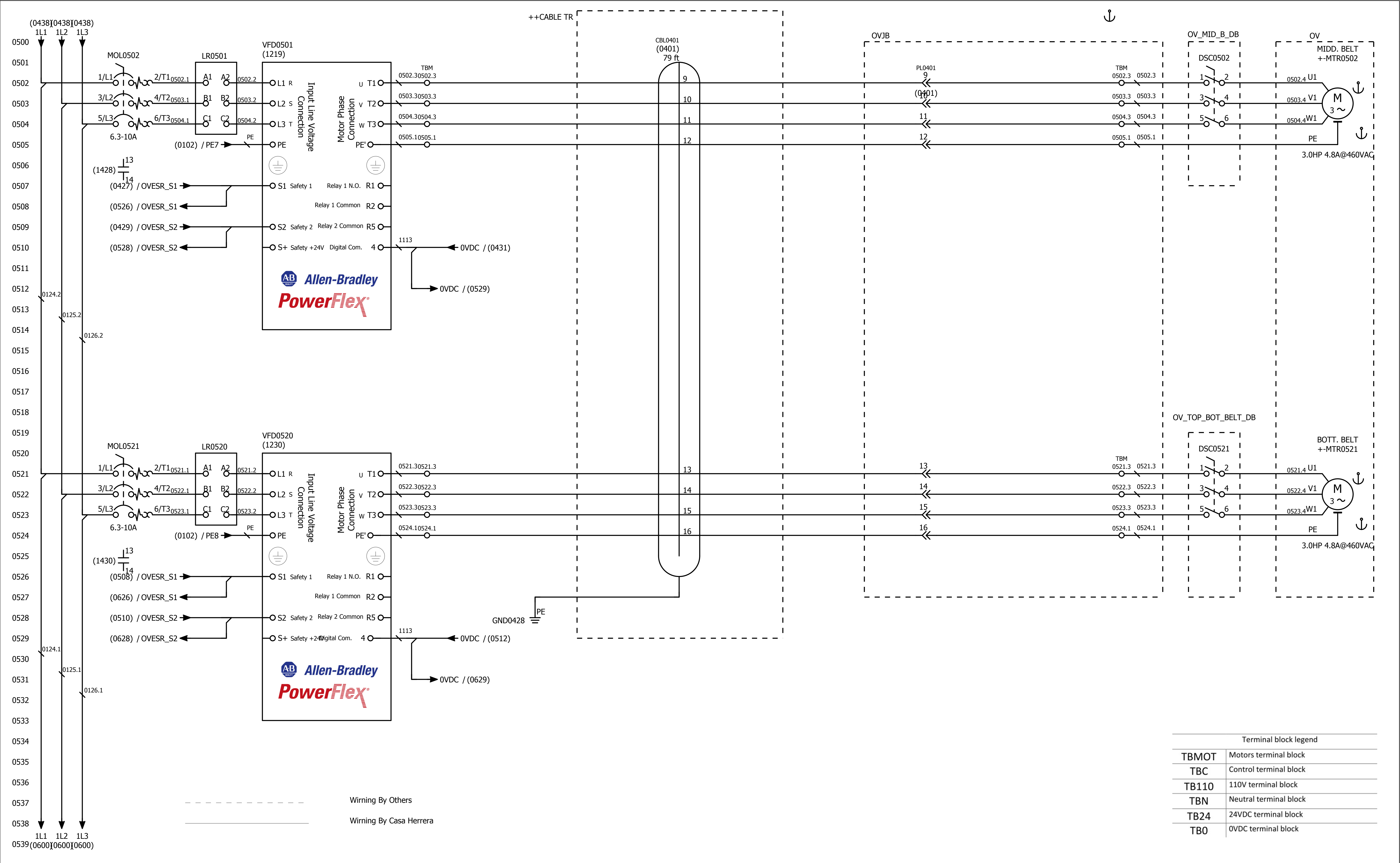
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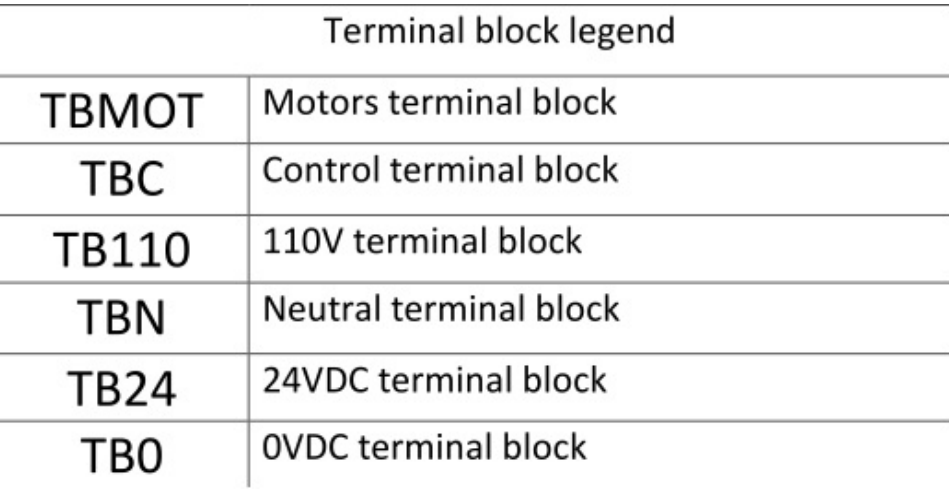


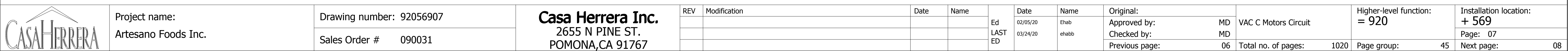




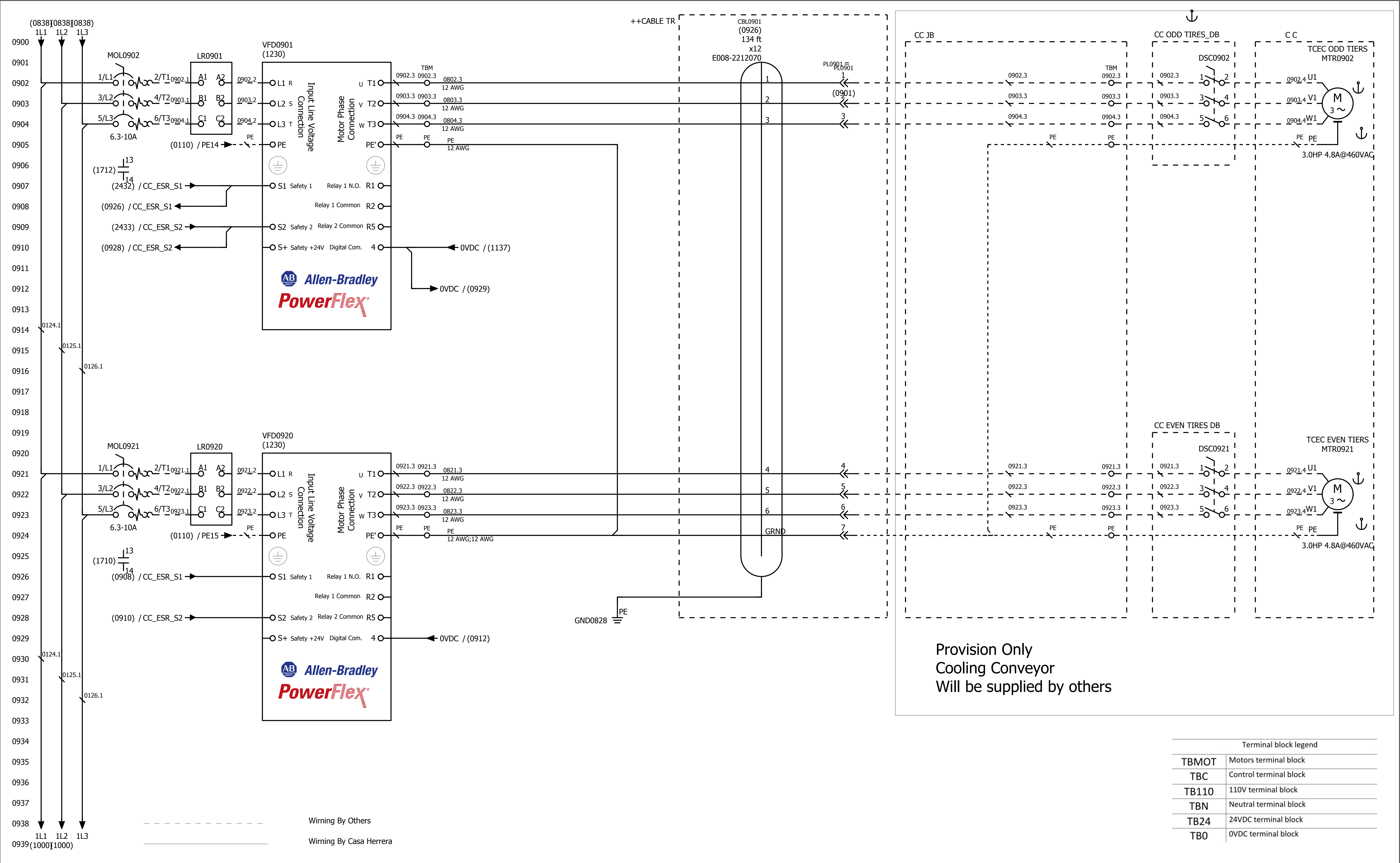




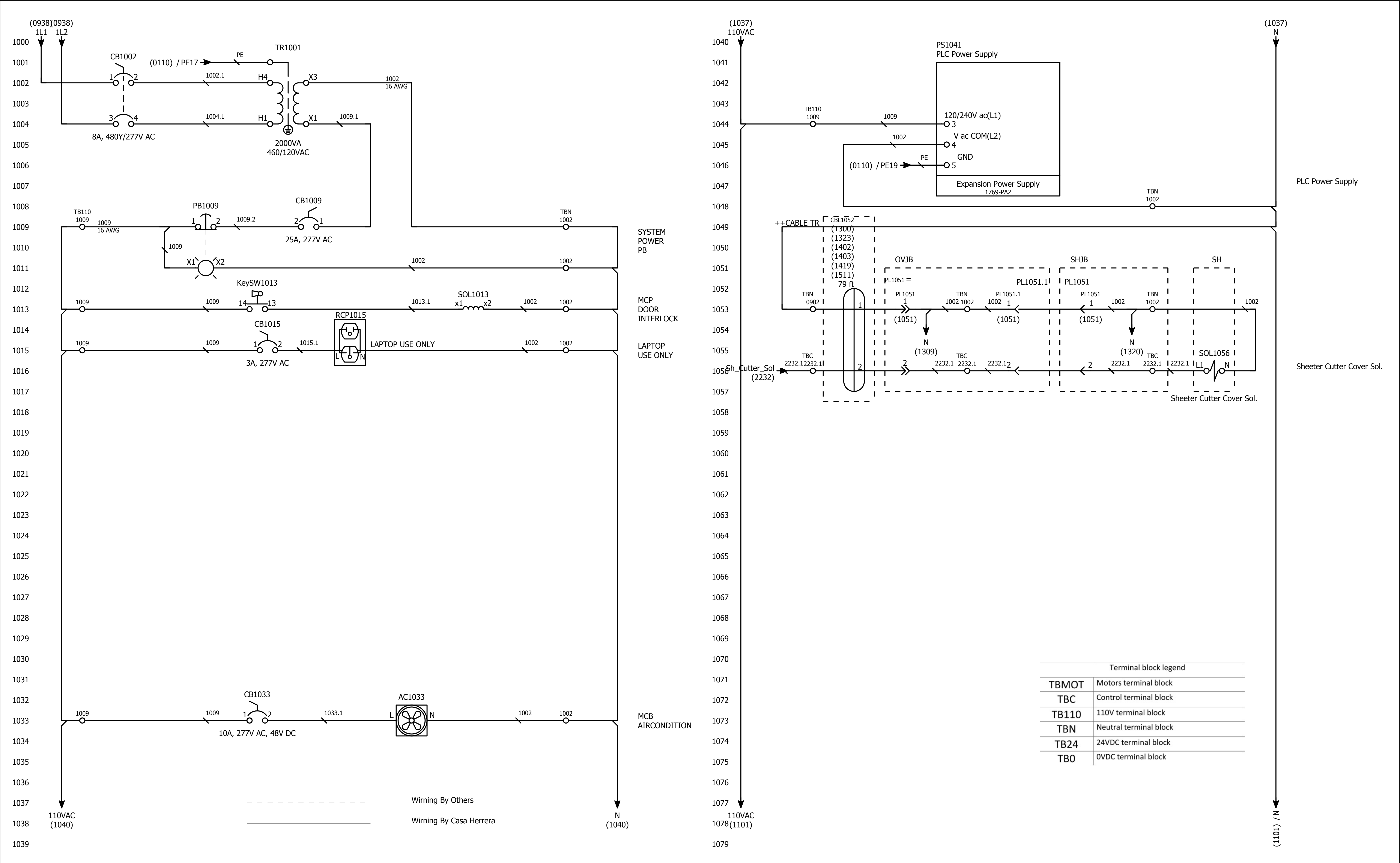




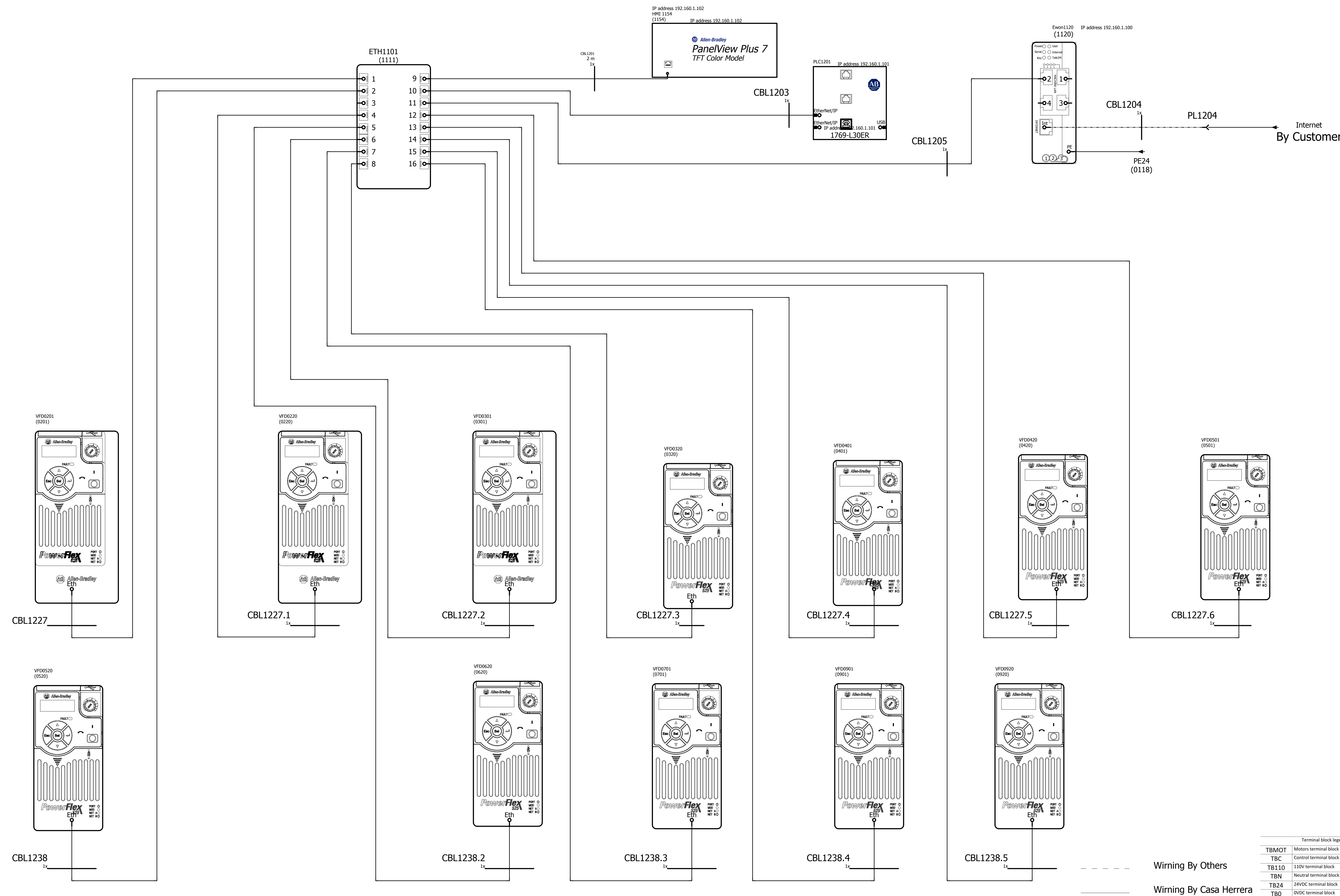




Terminal block legend	
TBMOT	Motors terminal block
TBC	Control terminal block
TB110	110V terminal block
TBN	Neutral terminal block
TB24	24VDC terminal block
TBO	0VDC terminal block



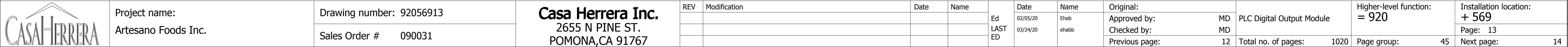
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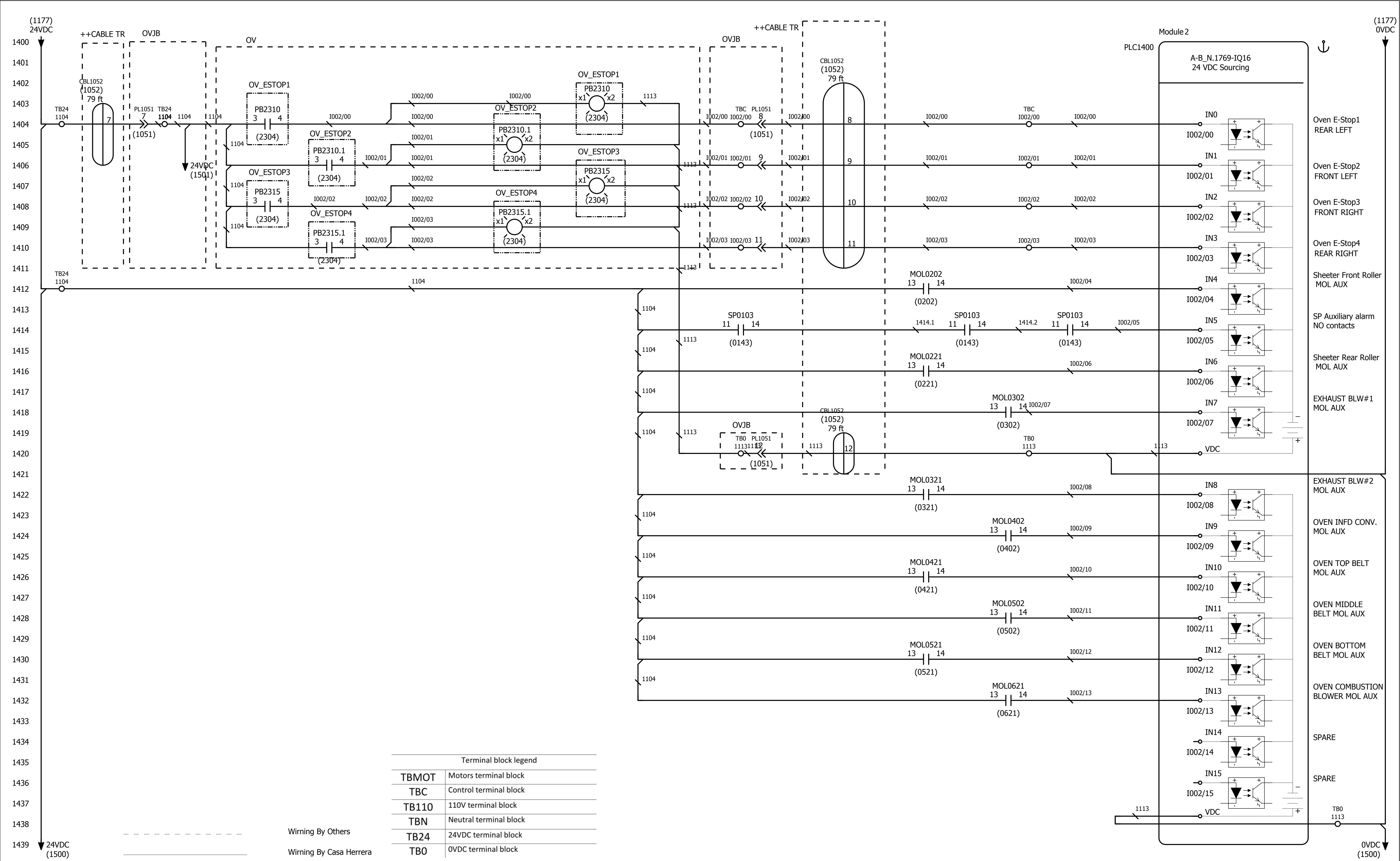


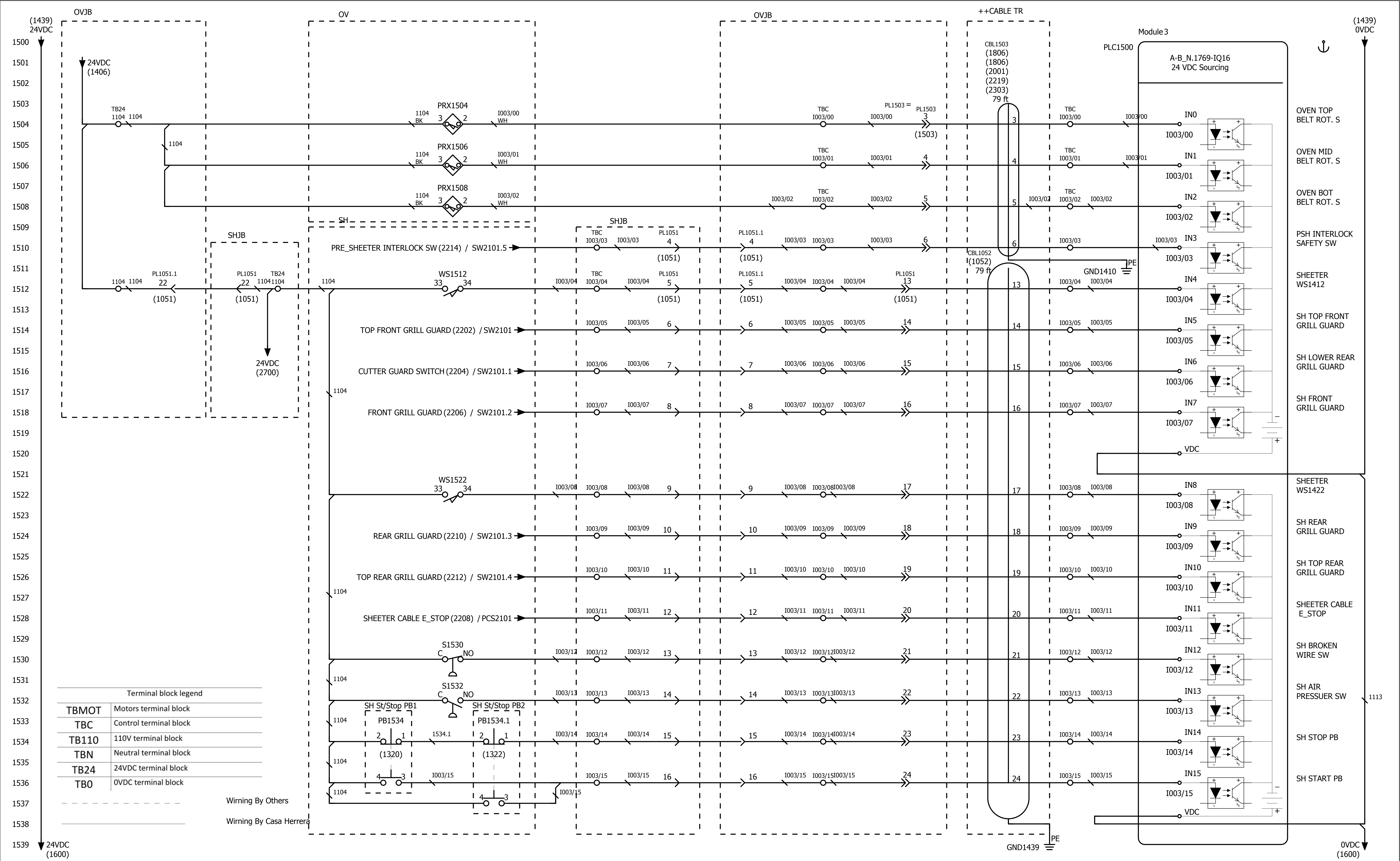
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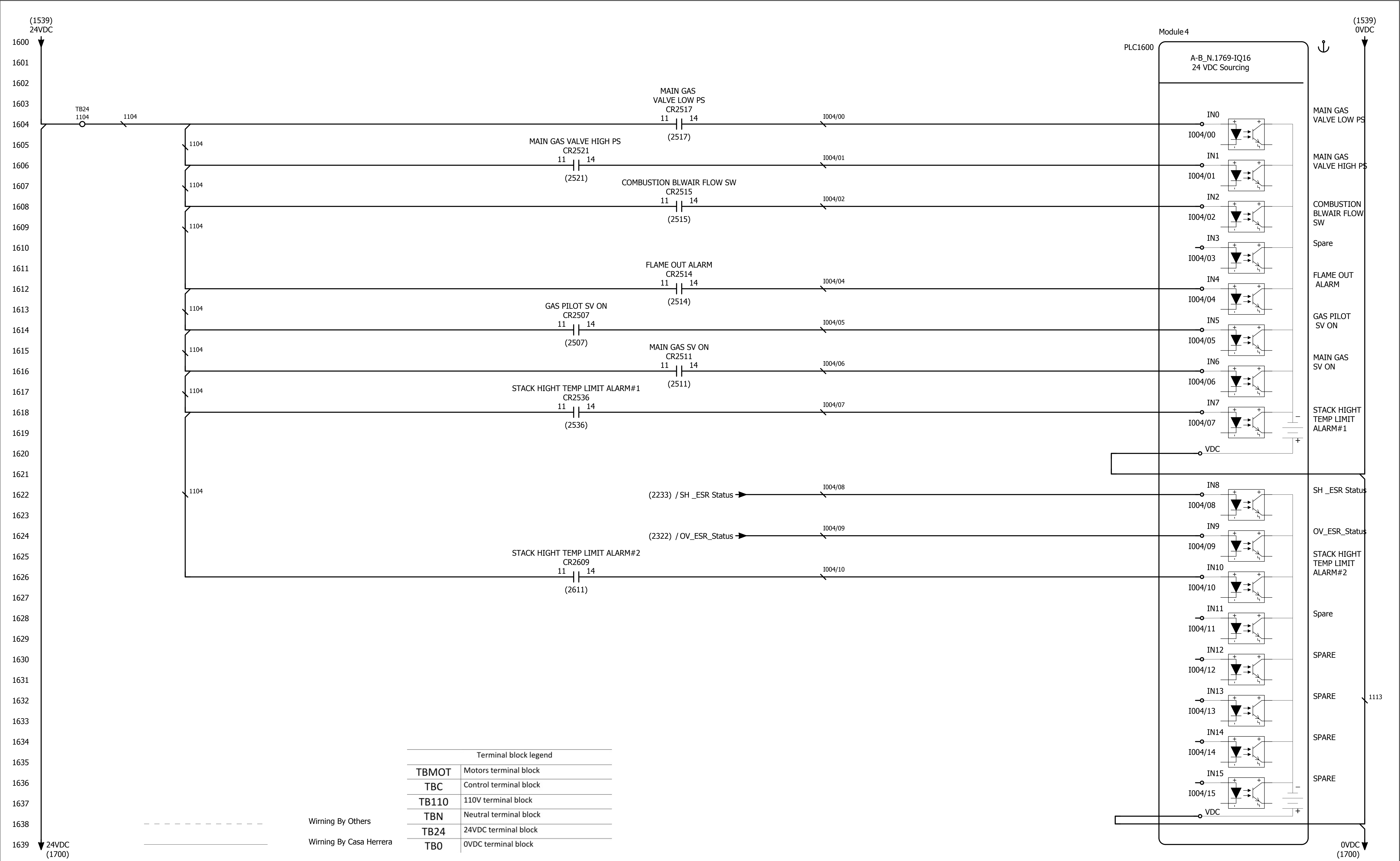
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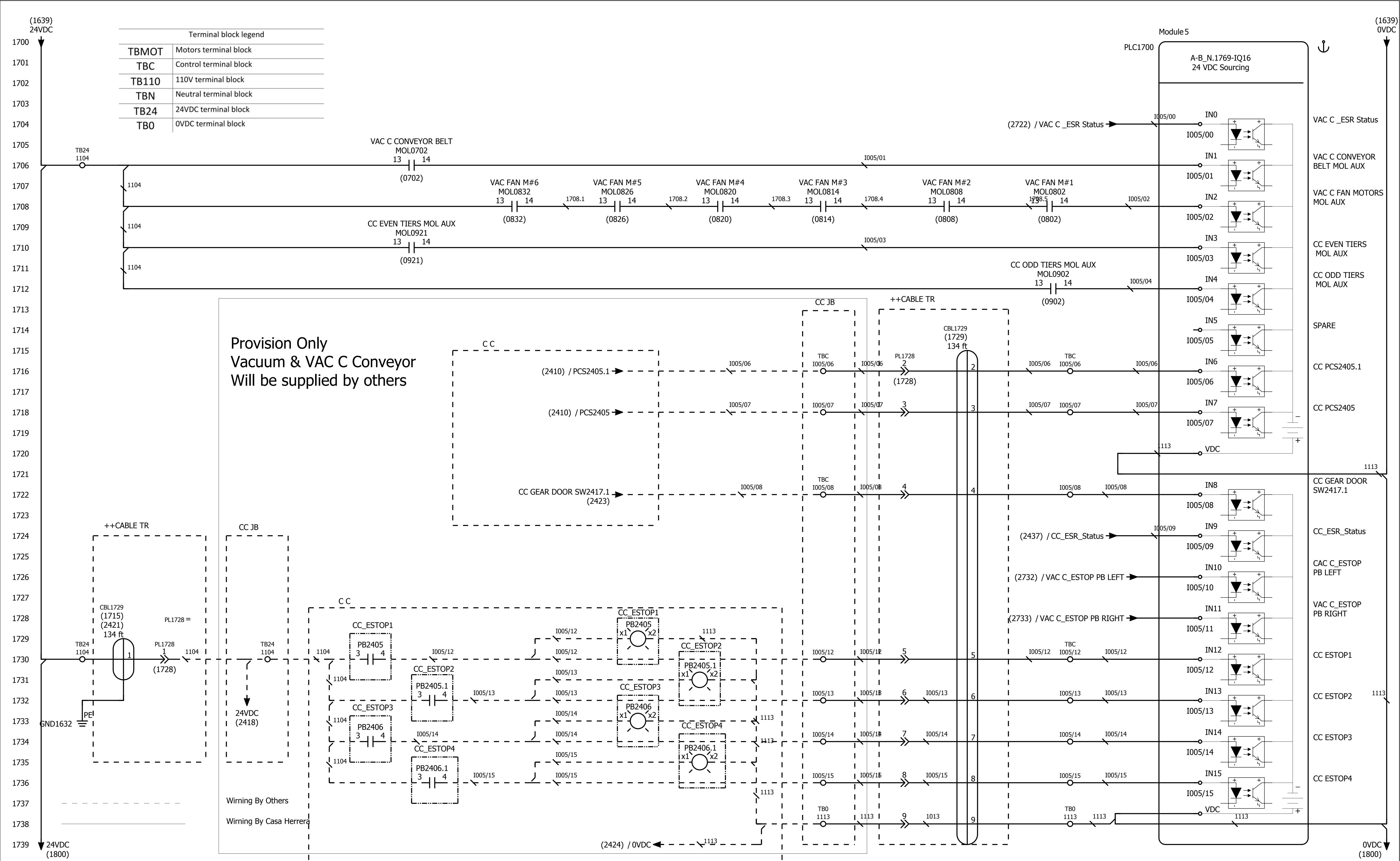
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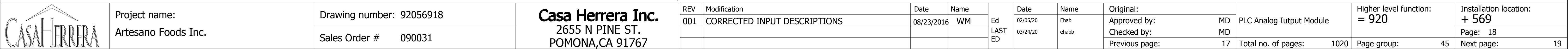




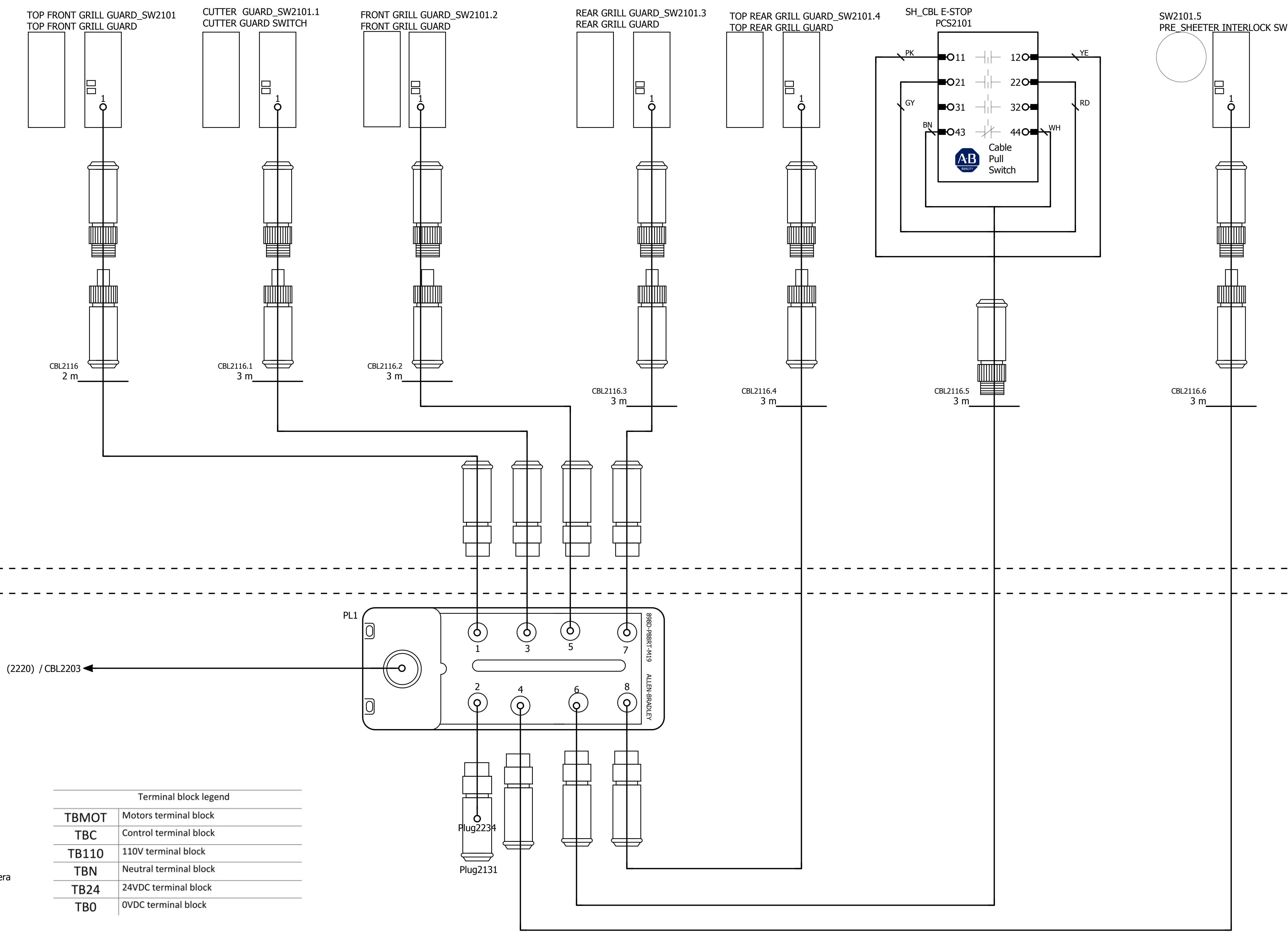


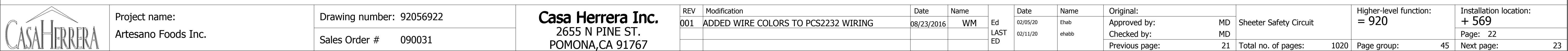


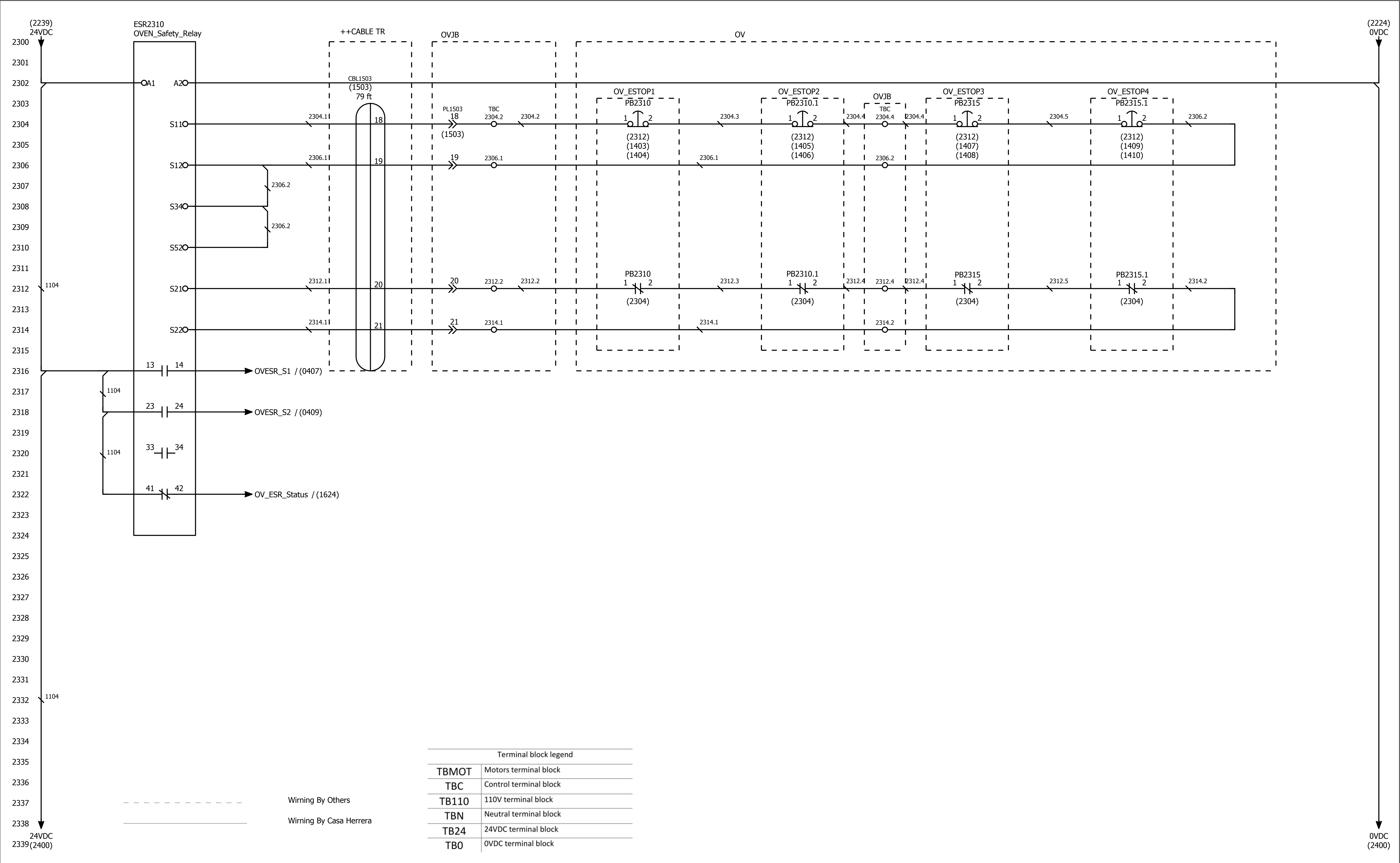


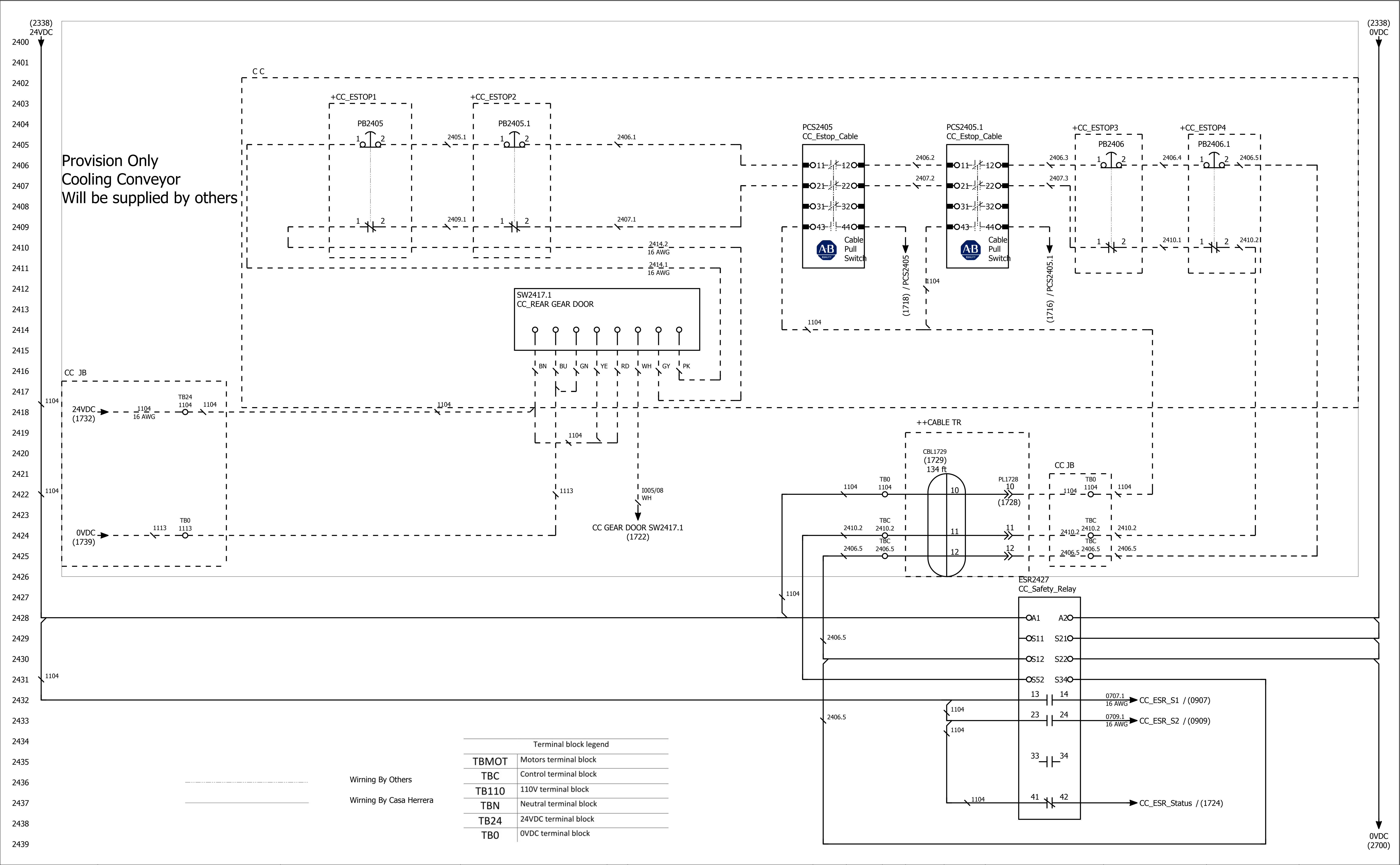


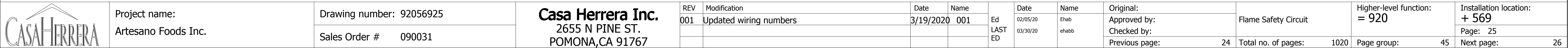
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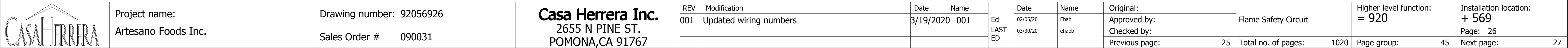


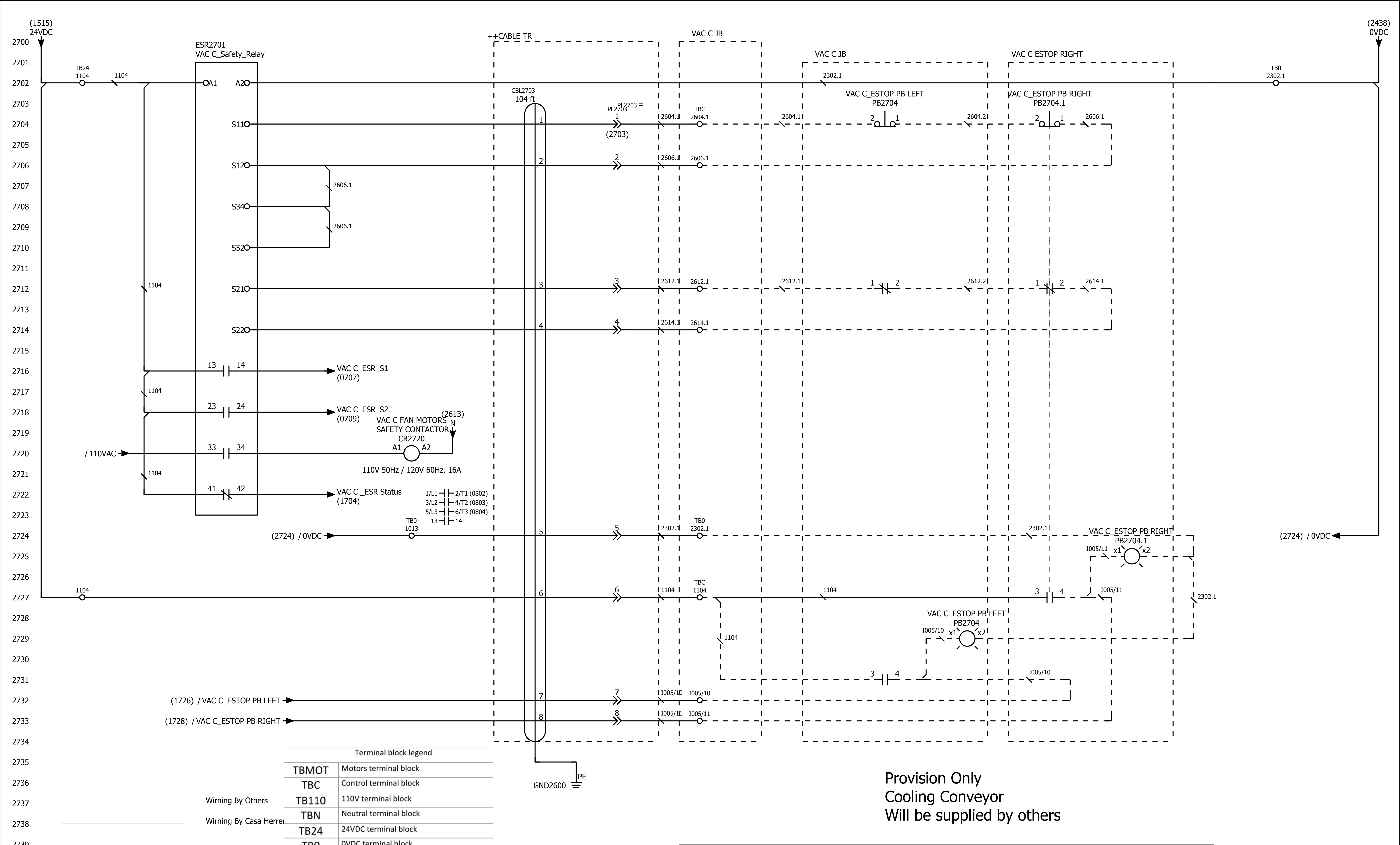










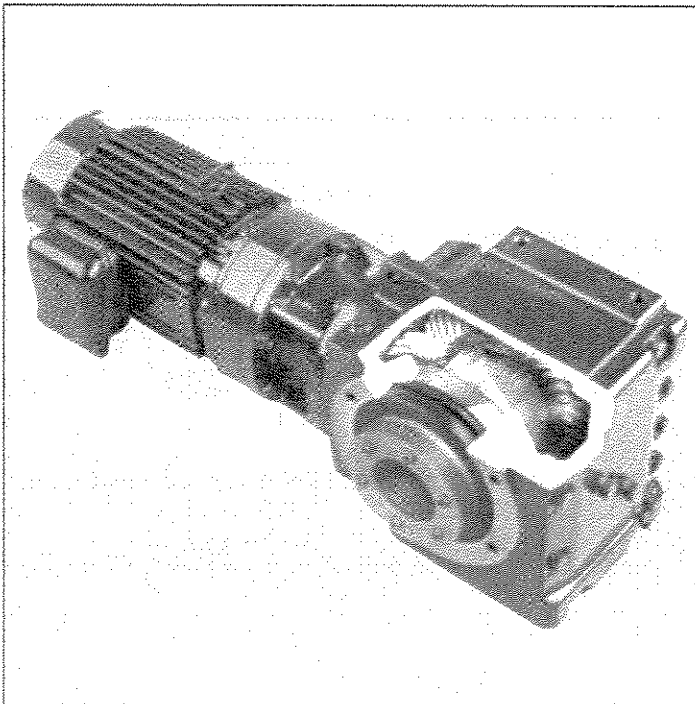


SM-BEVEL BUDDYBOX

Right Angle Cyclo

Operating and Maintenance Manual

NEW Shaft Mounted Right Angle Reducer
and Gearmotor



**THE
AVAILABLE
SOLUTION,
WORLDWIDE.**

 **SUMITOMO**
MACHINERY CORPORATION OF AMERICA

Manual
07.050.60.001

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MOUNTING SPECIFICATIONS

Table 1 - Permissible Shaft Tolerance

Shaft dia.	Tolerance
3/4" - 1 1/8"	+ 0 - .005"
1 3/16" - 2"	+ 0 - .006"
2 1/16" - 3 1/8"	+ 0 - .007"
3 3/16" - 4 3/4"	+ 0 - .008"

Table 3 - Stock Bushing Bores

Size	Inch Sizes	Metric Sizes	Min. Bore **
A	1 15/16, 2 3/16	50, 55	1 11/16
B	2 3/16, 2 7/16	60, 65	1 5/8
C	2 7/16, 2 15/16	65, 75	2 3/16
D	2 15/16, 3 7/16	75, 85	2 7/16
E	3 7/16, 3 15/16	90, 100	2 15/16

** Min. Bore is Also Stock But Needs Slitting.

Table 2 - Bushing Screw Tightening Torques

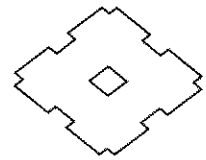
Size	Screw Size Qty. & Code	Screw Torque (lb. ft.)
A	6 X M12 942727	37.5
B	6 X M12 942727	37.5
C	6 X M16 942735	94
D	6 X M20 942740	180
E	6 X M20 942740	180

Table 4 - Bushing Bores Tolerances

Inch	Tolerance	Metric (mm)	Tolerance*
1 11/16 - 1 15/16	+ .003 / .001	40 - 50	.064 / .025
2 - 2 7/16	+ .003 / .001	50 - 65	.076 / .030
2 7/16 - 3 7/16	+ .003 / .001	65 - 80	.076 / .030
3 7/16 - 3 15/16	+ .003 / .001	80 - 100	.090 / .036

* Metric Tolerances are F8.

INSTALLATION



The Bevel Buddybox tapered bushing is designed to give easy on, easy off fit to the shaft for assembly and maintenance while giving a firm grip to the shaft in operation. The tapered bushing is fitted from the end of the hub; you should insure you have adequate access for correct tightening of the screws.

NOTE: Satisfactory performance depends on proper installation. Therefore all instructions on this page and in this manual must be carefully followed.

Fitting the Reducer on the Shaft (Bushing type)

1. Check the size and condition of the shaft to which the reducer will be fitted. Permissible shaft tolerances are given in Table 1 (previous page).
2. Ensure all mating surfaces of the hub, the inside and outside diameters of the tapered bushing and the shaft are free from burrs and corrosion. Clean each surface with a solvent to remove all traces of grease and oil.
3. Lightly oil each screw and insert into the bushing flange; ensure they do not protrude beyond the rear face. See Diagram 1.
4. Insert the face keys of the thrust collar into the slots on hub and center. Screw the tapered bushing into the hub in a clockwise direction until the flange contacts the thrust collar. See Diagram 2.
5. Unscrew the tapered bushing until the screws are aligned with the pockets in the face of the thrust collar and a gap of 1 mm minimum exists between the flange and thrust collar. This may be easier if one of the screws are removed and the pocket viewed through the threaded hole. Finger tighten all screws into the pocket. See Diagram 3.
6. Slide the reducer onto the shaft at least as far as the counter bore in the taper grip bushing. See Diagram 4. Be sure that bushing and shaft are aligned and gearbox weight is supported for correct mounting alignment. Gradually tighten each screw in a star pattern to the torque levels shown in Table 2. A torque wrench is required.

CAUTION: Slippage of the tapered bushing can be avoided by proper shaft and bushing selection and correct pretensioning of the bushing locking screws. If repeated slippage occurs, bushing disassembly and reducer removal may be difficult.

7. Grease fill the cavity between the bushing counterbore and shaft to prevent a build-up of corrosion on the shaft end which may affect subsequent removal of the reducer. This is to be done after assembly of the unit. See Diagram 5.
9. Install the torque arm assembly if one is used. See Installing the Torque Arm.

NOTE: Be sure to use a torque arm or other device to prevent rotation when mounting by shaft only.

10. After mounting is complete, the Bevel Buddybox can then be filled with oil. Please follow proper guidelines for oil lubrication. Grease lubricated units are prefilled at the factory. Refer to lubrication information.
11. After the reducer has been running for 20 to 30 hours retorque the screws to the values in Table 2 using a torque wrench. Screw torques should be checked at normal service intervals (every 6 months).

DIAGRAM 1

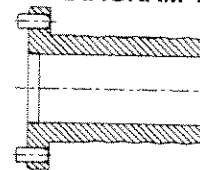


DIAGRAM 2

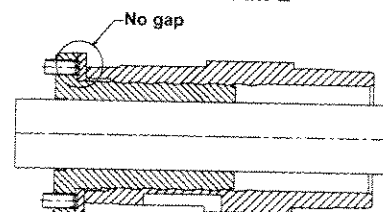


DIAGRAM 3

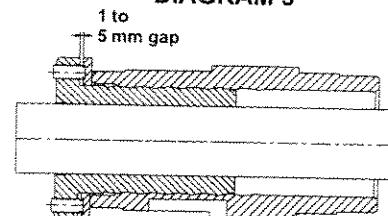


DIAGRAM 4

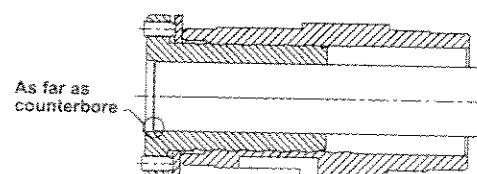
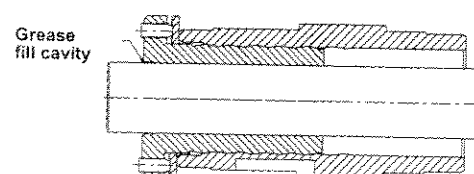


DIAGRAM 5



INSTALLATION

Fitting the Reducer on the Shaft (Keyed Hollowshaft type)

Mounting procedure:

1. Smear the surface of the shaft (e) with molybdenum disulfide compound.
2. Turn nut (b) and slide the reducer over the driven shaft. Use ring (c) as necessary.
3. After mounting the reducer on the shaft, install bolt (f). See Diagram 7.

NOTE: The bore should be protected by a cover (g).

4. If the driven shaft does not have a shoulder, a spacer (h) should be used. See Diagram 8.

Removal procedure:

1. Remove ring (d), mount bolt (n) and reset ring (d). Attach bolt (j) to ring (d) and turn bolt (j) to disconnect the hollow shaft from the driven shaft.

NOTE: Parts a through j and n are not provided by Sumitomo.

DIAGRAM 6

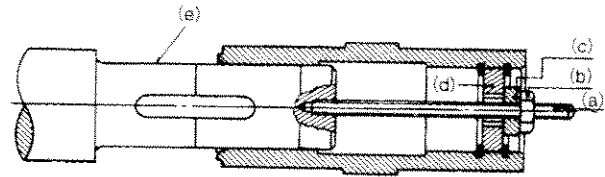


DIAGRAM 7

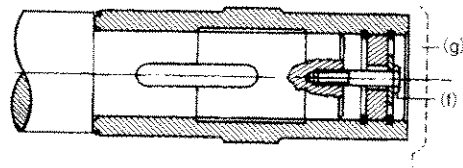


DIAGRAM 8

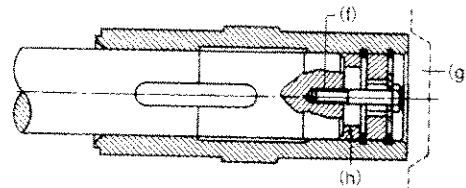
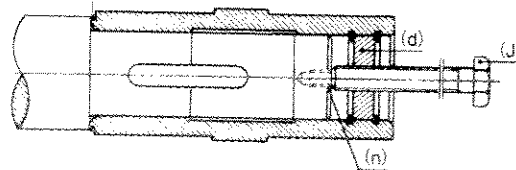
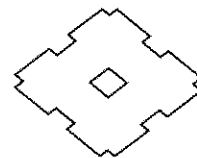


DIAGRAM 9



INSTALLATION



Installing the Torque Arm

1. Remove the cotter pins or nuts (6) from the bolts (5) on the torque arm.
2. Insert the bolts through the torque arm clevis and ear on the housing (3) and mount (1). Reinstall cotter pins or nuts (6).
3. Adjust the length of the torque arm by loosening the locknuts (4), turning the body of the torque arm (2), and retightening the locknuts (4).
4. Install the torque arm mount (1) on a rigid support so that the torque arm will be at $90^\circ (\pm 30^\circ)$ to a line between the mounting hole and the output shaft, to ensure the torque arm will be loaded in tension.
5. The torque arm should be attached to a firm mounting position with the torque arm mount (1) provided or a suitable substitute. The attachment points of each end of the torque arm (gearcase and bracket) should be on the same plane on centerline when viewed from the side. See Diagram 6.
6. The torque arm should be attached and tightened while maintaining correct gearcase position (top, bottom, side, etc.). The torque arm turnbuckle should be hand tightened to avoid overtightening.
7. If an audible sound of metal to metal contact is heard upon start-up of the drive it is likely one or both of the following:
 - a) Torque arm setting up due to some slack in the linkage. This is normal.
 - b) The output gears in the Buddybox contacting due to backlash in the gear set. This is also normal.

Note: Whenever possible, the torque arm should be installed in tension not in compression.

8. For shaft mounted units the reducer will tend to rotate in the direction opposite that of the output shaft (see arrows). The torque arm will prevent this motion thus being in tension.

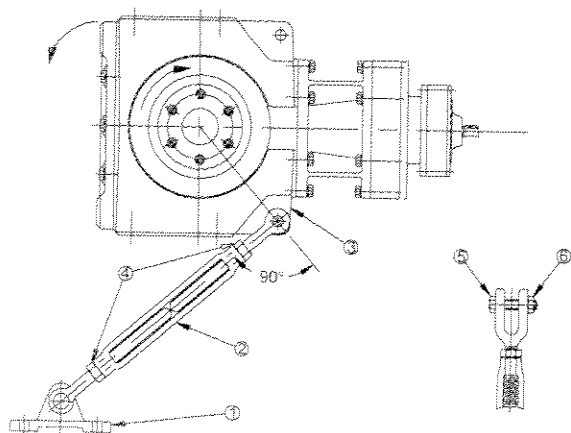
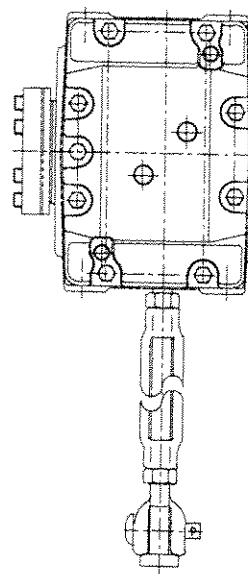


DIAGRAM 6



Removing the Reducer from the Shaft

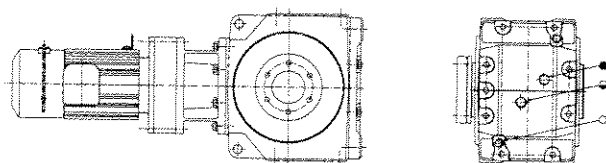
1. Ensure that the reducer is supported and cannot fall free when the grip of the tapered bushing is released.
2. Release each screw gradually until they are free from the pockets of the thrust collar.
3. If the tapered bushing does not self release, give the flange a sharp tap with a brass or copper mallet to break free the taper. This will free the reducer from the shaft. Be careful not to damage the tapped holes.
4. Hand tighten two screws against the thrust collar before attempting to slide the reducer off the shaft. This will stop the hub from pulling up the opposite taper flanks of the tapered bushing and prevent it from relocking on the shaft as the reducer is removed.
5. Slide the reducer from the shaft. If corrosion or paint prevents the reducer from sliding off the shaft, it is possible to utilize two of the screw holes to secure a puller plate to the bushing.

LUBRICATION

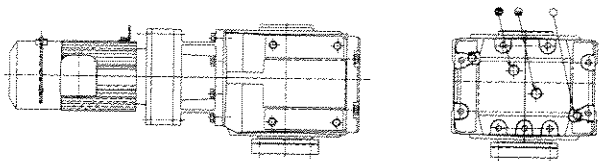
Bevel Buddybox reducers and gearmotors are shipped without oil. Before running they should be filled with the appropriate amount of the correct lubricant as shown in the tables, dependent on the mounting position-see table below. Please consult the factory for additional information not covered here.

Mounting Positions

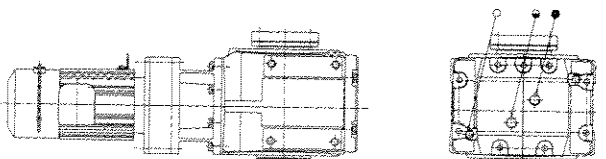
- A & B sizes are grease packed before leaving the factory.
- When filling the unit with oil, tilt the unit approximately 5° to allow oil to flow more quickly to the Cyclo portion and continue to fill the unit until oil flows out from the overflow hole. Remove the 5° tilt from the unit and allow unit to sit for a few minutes until oil no longer flows out of the overflow hole.
- If the unit purchased is oil lubricated in both the Cyclo and the gearcase, the oil is put into the gearcase and flows into both portions.
- Before reinstalling the plugs, reapply Teflon tape to the threads to prevent leakage.



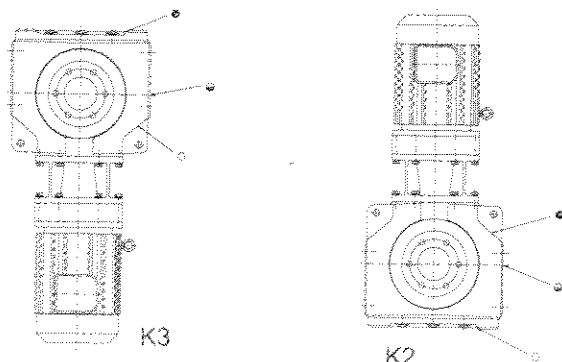
K1, R1



K6



K5



K3

K2

Recommended Oil for SM Bevel Buddybox & Cyclo

Ambient Temp.	14°F-32°F (-10°C-0°C)	32°F-95°F (0°C-35°C)	95°F-122°F (35°C-50°C)
Viscosity @ 40°C (104°F) cST	61.2 - 74.8	90 - 165	198 - 506
ISO Viscosity Grade	68	100 - 150	220 - 460
AGMA Viscosity Grade	2EP	3EP 4EP	5EP 7 EP
Viscosity @ 100°F (38°C) SSU	284 - 347	417 - 765	916 - 2719

Oil Quantity (In gallons)

Bevel Buddybox size	Oil Quantity - gear			Oil Quantity - Cyclo	
	Pos. K1	Pos. K5	Pos. K6	All Pos.	Pos. K2,K3
A	G	G	G	G	G
B	G	G	G	G	G
C	1.4	1.1	1.1	*	G
D	2.2	2.1	2.1	*	G
E	4.6	3.8	3.4	*	G

*Oil flows from the bevel gearcase into the Cyclo portion.

Oil Replenishment and Change Interval

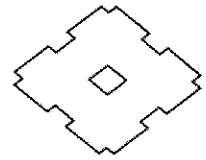
- Maintain proper oil levels at all times.
- Check oil quantity every 2500 hours. If the oil is contaminated, burned or waxed, change the oil immediately and flush if necessary.
- Under normal operating conditions, Sumitomo recommends an oil change every 5000 hours of operation. The intervals should not exceed 2 years.
- An oil change after the first 500 hours of operation is highly recommended.
- If the unit will be running in a high ambient, high humidity, or corrosive environment, the lubricant will have to be changed more frequently. Consult the factory for recommendations.

Grease Replenishment and Change Interval

- Grease replenishment is normally not necessary. If Cyclo portion is a double reduction, please see the next page for recommendations.
- Drain, flush and regrease the bevel gearcase every 3000 to 5000 hours of operation.
- There should be an overhaul of the unit every 2 to 3 years.

- Oil Fill Plug
- Oil Overflow Plug (Oil Level)
- Drain Plug

LUBRICATION



Designated Greases

For additional information, please refer to Lubrication Specification Sheet No. 03.301.63.001.

Ambient Temperature F° (C°)	Single Reduction (Maintenance Free)	Double Reduction	Bevel Gear Section
-5 ~ 122 (-15 ~ 50)	NLGI No. 2	NLGI No. 2	Shell Alvania EP RΦΦ

Grease Replenishment and Change Interval (Cyclo)

Model	Condition		Interval
Single & Double Reduction (Maintenance Free Type)	Replenishment		NOT REQUIRED
	*Overhaul		Every 20,000 Hours or Every 4 ~ 5 Years
Double Reduction (Non-Maintenance Free Type)	Replenishment	Less Than 10 Hours Per Day Operation	Every 3 ~ 6 Months
		10 ~ 24 Hours Per Day	Every 500 ~ 1000 Hours
	Change	Speed Reduction Mechanism, High Speed Shaft Bearings (Speed Reducer Type)	Every 2 ~ 3 Years
		Slow Speed Shaft Bearings	Every 3 ~ 5 Years

*Overhauling consists of disassembling the unit, replacing the seals and gaskets, cleaning the internal parts, and then repacking the unit with designated grease.

Note 1: Frame sizes 4100-4125 are maintenance free units. Grease replenishment is not necessary. Where longer life of the drive is expected or if relubricating is preferred before recommended period of time, refer to Tables 6, 7 and 8.

Quantities of Grease (Ounces)

Frame Size	4085	4090 4095 4097	4100 4105 410H	4110 4115 4125
Speed Reduction Mechanism	.7	2.1	3.5	8.5
Slow Speed Shaft Bearings	.9	2.3	3.9	4.9

Frame Size	4105DA	4115DB	4135DC	4145DB	4165DC	4175DC
Speed Reduction Mechanism (First Stage)	0.7	2.1	2.3	1.4	5.3	
Speed Reduction Mechanism (Second Stage)	3.5	8.5	15.9	15.9	26.5	35.3
Slow Speed Shaft Bearing (Second Stage)	3.9	4.9	10.6	10.6	10.6	17.6

Note 1: Replenish grease to the reduction mechanism 1/3 to 1/2 of quantities for the first reduction stage described in Table 8 in accordance with replenishment interval recommended in Table 7.

Note 2: When the unit is disassembled for overhauling, refill with grease in quantities indicated in Table 8. Or alternatively, 80% of the space around reduction mechanism and slow speed shaft bearings of single reduction units and 50% around reduction mechanism of both first and second stage of double reduction units.

Slightly larger quantities may be supplied to lower reduction ratio units, and somewhat smaller quantities for high reduction ratio units.

Apply grease liberally to the central part (i.e., around the eccentric bearings) of the mechanism. Apply grease to both the slow speed and high speed shaft bearings as you would do to ordinary bearings at time of re-assembly.

Note 3: If excessive grease is added, agitation heating of the grease will raise the operating temperature of the unit. Avoid excessive greasing, however, as the reverse case, when the grease is insufficient it will raise the operating temperature due to the breakdown of the lubrication films on the eccentric bearing.

If a rise in the operating temperature is found supply grease immediately.

BEVEL GEAR CASE

Replacement of Parts

NOTE: Using standard tools, the reducer can be dismantled and reassembled. Cleanliness is most important to prevent the entry of dirt into the bearings and other parts of the reducer. A tank of clean solvent, an arbor press and equipment for heating gears and bearings should be available for shrink fitting these parts onto the shafts.

Procedure

1. Wash off all removable dirt from the assembly.
2. Drain unit of lubricant.
3. Remove the Cyclo unit from the input cartridge (16).
4. Remove the cartridge bolts (13) and input cartridge (16).
5. Remove the taper grip bushing (29) from the output hub (5).
6. Remove the end cover bolts (27) and end cover (22) from the housing.
7. The output assembly may be stripped as required using snap-ring pliers and a press.
8. The cartridge assembly may be stripped in the following way:
 - Remove bolt (6) and pinion thrust plate (7) from top of pinion. It may be necessary to heat the bolt prior to removal due to the use of an adhesive thread locker during assembly.
 - Remove bevel pinion (11) and shim (19).
 - Support the input cartridge (16) on a press table, and using suitable protection for the input shaft (17), press

it from the input cartridge (16). Care must be taken to prevent the shaft assembly from dropping during this procedure.

- To remove the taper roller bearing (12) and input collar (18), support under the taper roller bearing (12) and press the shaft through the bearing and collar. Care must be taken to prevent the shaft from dropping during this procedure.
- Keep input shim (19) in front of taper roller bearing (12).

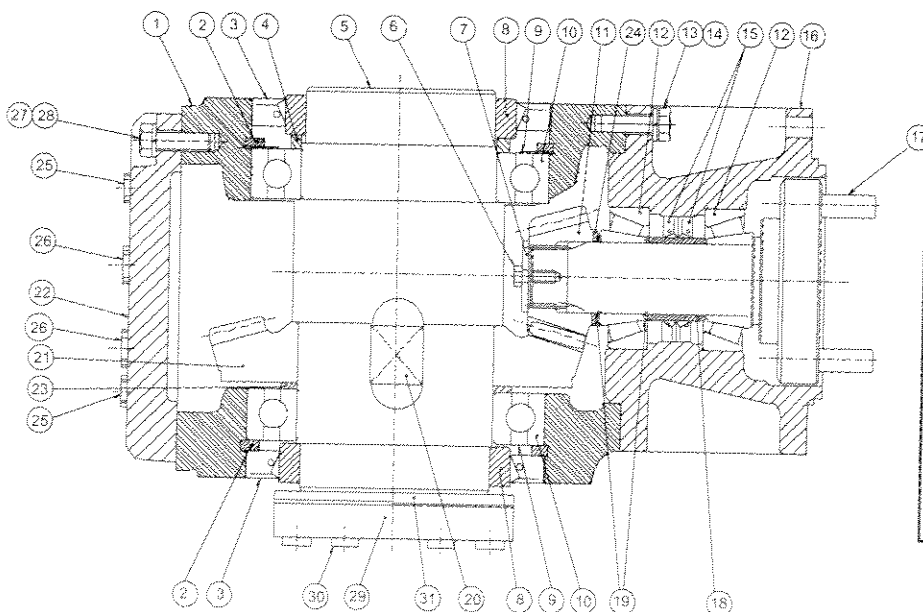
NOTE: Lip-type oil seals are used. Great care should be taken during dismantling and reassembly to avoid damage to the rubbing surfaces. Any burrs or damage to these surfaces must be carefully removed. When seals are removed, they should not be reused.

Ordering Parts

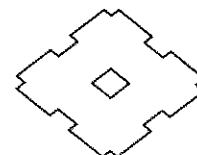
NOTE: When ordering parts for a reducer, the following should be specified:

1. The reducer size, serial number, part code number description and quantity required.
2. The part code number can be found on the "Parts Code Numbers" page.
3. When a bevel pinion or bevel gear is replaced, the mating gear or pinion must also be replaced. It is recommended that the complete bevel assembly (output assembly and cartridge assembly) be replaced. This will ensure that proper bevel gear tooth contact and backlash is achieved.
4. For parts replacement or other information regarding the Cyclo portion of your unit, please use the Cyclo reducer and gearmotor section of this manual.

BEVEL GEAR OUTPUT SECTION



GENERAL CONSTRUCTION-PARTS LIST



Parts Code Numbers

REF	DESCRIPTION	QTY	A CODE	B CODE	C CODE	D CODE	E CODE
1	HOUSING	1	TDMD0419	TDMD0421	TDMD0423	TDMD0425	THMD0427
2	RETAINING RING	2	241K130-NG	241K150-NG	241K170-NG	241K200-NG	241K320-NG
3	OIL SEAL	2	531N10013012G	531N11515012G	531N13517012G	531N16020012G	531N19023016-NG
4	SPACER	1	TRMA1728-1	TRMA1728-2	TRMA1728-3	TRMA1728-4	TRMA1728-5
5	OUTPUT HUB	1	TLMC1124	TLMC1125	TLMC1126	TLMC1127	TLMC1128
6	HEX HEAD BOLT	1	001H008A030NG	001M010A040NG	001M016R060NG	001M020R070NG	—
7	PINION THRUST PLATE	1	TMMA1727-1	TMMA1727-2	TMMA1727-3	TMMA1727-4	CP10810
8	OUTPUT COLLAR	2	TRMA1731	TRMA1732	TRMA1733	TRMA1734	TRMA1735
9	OUTPUT SHIM	2	U215052000	U215054000	U215056000	U215059000	U215063000
10	BALL BEARING**	2	5006017-G	5006020-G	5006022-G	5006026-G	5006030-G
11	BEVEL PINION	1	2008456	2008458	2008460	2008462	2008464
12	TAPER ROLLER BEARING	2	50032206-G	5004T30307D-G	50030311D-G	50033213-G	50032314-G
13	HEX HEAD BOLT	*	000H008A025NG	000H010A035NG	000H010A035NG	000H012A045NG	000H012A045NG
14	LOCKWASHER	*	062S008-NG	062S0A0-NG	062S010-NG	062S012-NG	062S012-NG
15	OIL SEAL**	2	530N405565-G	530N45658-G	—	—	—
16	INPUT CARTRIDGE	1	TDMD0429	TDMD0431	TDMD0433	TDMD0435	TDMC1671
17	INPUT SHAFT	1	THMD0485	THMD0486	THMD0487	THMD0488	THMD0689
18	INPUT COLLAR	1	TRMA1755	TRMA1756	TRMA1752	TRMA1753	TRMA1754
19	INPUT SHIM	2	TMMA1741	TMMA1737	TMMA1740	TMMA1742	TMMA1744
20	OUTPUT KEY	1	TMMA1766-1	TMMA1766-2	TMMA1766-3	TMMA1766-4	TMMA1730-5
21	BEVEL GEAR	1	2008457	2008459	2008461	2008463	2008465
22	ENDCOVER	1	TAMC1099	TAMC1100	TAMC1101	TAMC1102	TAMC1103
23	SPACER	1	TRMA1729-1	TRMA1729-2	TRMA1729-3	TRMA1729-4	TRMA1729-5
24	INPUT SPACER	1	—	—	TRMA1746	TRMA1747	TRMA1748
25	HEX SOCKET HEAD PLUG	2	343F004B-NG	343F004B-NG	343F006B-NG	343F006B-NG	343F006B-NG
26	HEX SOCKET HEAD PLUG	2	343F006B-NG	343F006B-NG	343F006B-NG	343F006B-NG	343F006B-NG
27	HEX HEAD BOLT	1	000H008A025NG	000H008A025NG	000H010A030NG	000H012A040NG	000H012A040NG
28	LOCKWASHER	*	062S008-NG	062S008-NG	062S010-NG	062S012-NG	062S012-NG
29	TAPER GRIP BUSH	1	AS REQ	AS REQ	AS REQ	AS REQ	AS REQ
30	BUSH SCREW	6	0942881	0942881	0942890	0942900	0942900
31	THRUST PLATE	1	016E6146	016F6146	016G6146	016H6146	016I6146
32	TORQUE ARM ASSEMBLY	1	988TABBB-AG	988TABBB-BG	988TABBB-CG	988TABBB-DG	988TABBB-EG

* Quantity varies with the size of the unit.

** These parts must vary depending on the mounting position. Please state full unit part number and serial number when ordering replacement parts.

GENERAL CONSTRUCTION – PARTS LIST

Cyclo Reducer Input Section

Fig. 1 Single Reduction

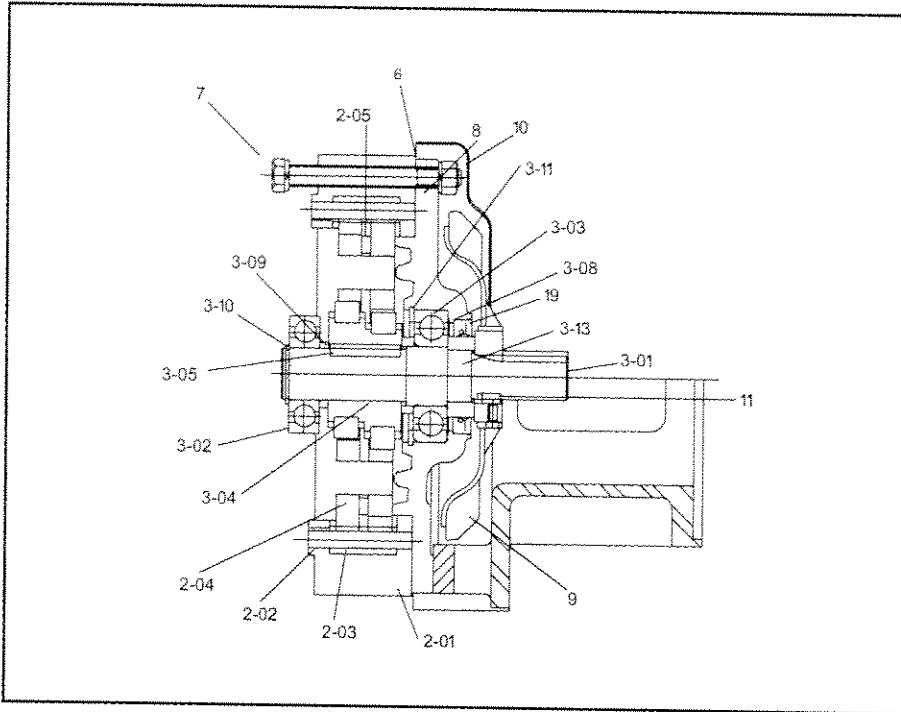


Fig. 2 Double Reduction

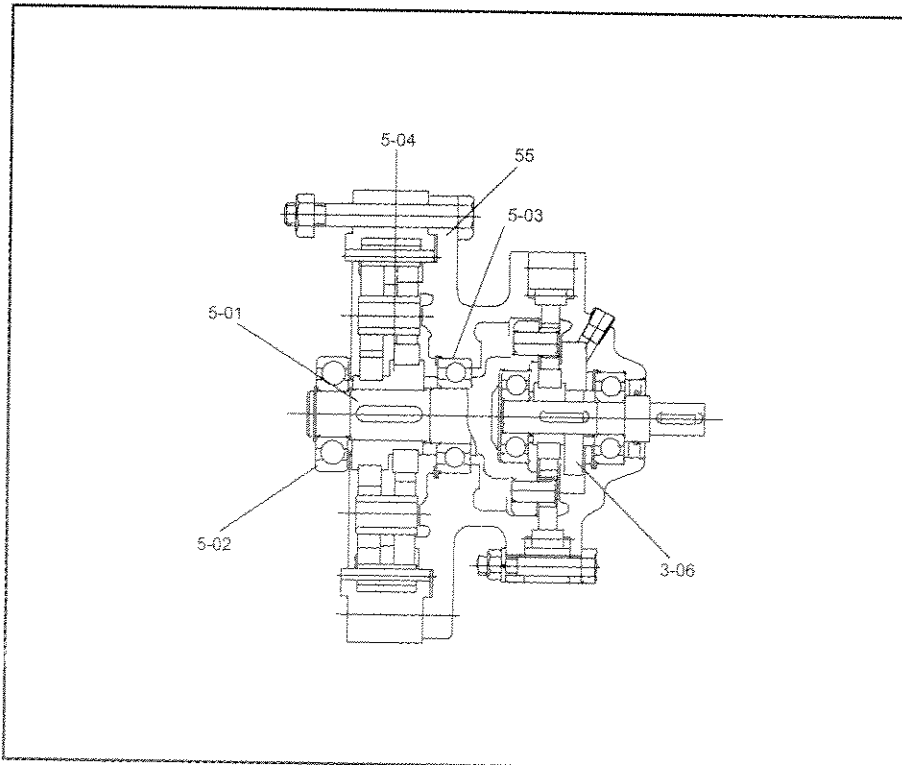


Table 1 Main Parts

Part No.	Part Name
2-01	Ring Gear Housing
2-02	Ring Gear Pins
2-03	Ring Gear Rollers
2-04	Cyclo Disc
2-05	Spacer Ring
3-01	High Speed Shaft
3-02	Bearing C
3-03	Bearing D
3-04	Eccentric CAM Assembly
3-05	Eccentric Key
3-06	Balance Weight*
3-08	Spacer
3-09	Spacer
3-10	Retaining Ring
3-11	Retaining Ring
3-13	Collar
5-01	Intermediate Shaft w/Pins
5-02	Bearing F
5-03	Bearing G
5-04	Eccentric CAM Assembly
6	Gasket Set
7	Casing Nuts & Bolts
8	High Speed End Shield
9	Cooling Fan & Set Screw
10	Fan Cover
11	Fan Key
19	High Speed Input Oil Seal
55	Intermediate Cover

*Single disc unit only.

BEARINGS, OIL SEALS, GASKETS

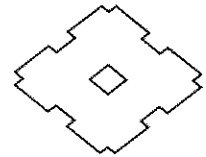


Figure 9

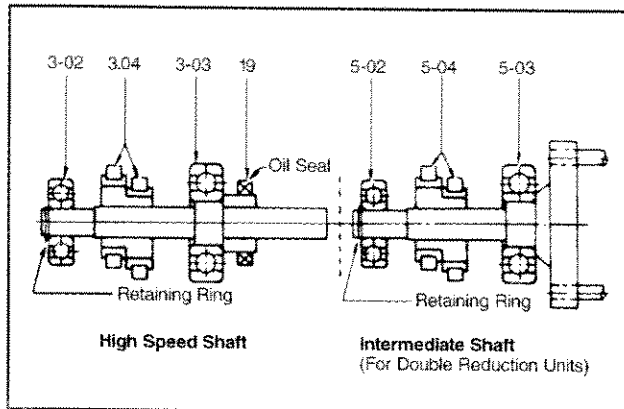


Figure 10

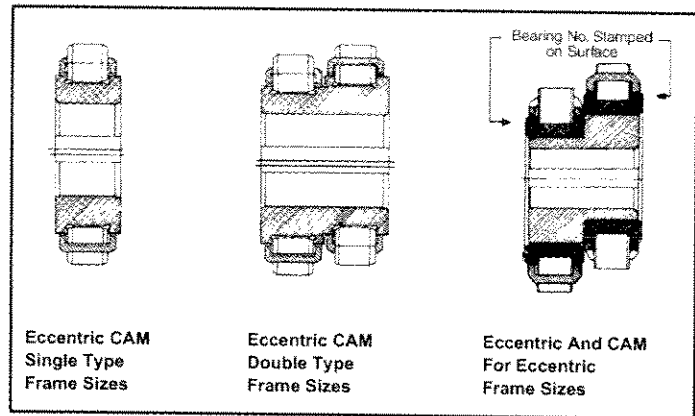


Table 19 Single Reduction Units

Frame Size	BEARINGS				OIL SEAL			
	HIGH SPEED SHAFT				HIGH SPEED SHAFT			
	Part No. 3-02	Part No. 3-04	Part No. 3-03	Part No. 19				
	Number	Number	Qty.	Number	Reducer	Type	Dimension	Qty.
4085	6301	19UZS208T2	1	6301Z	S		17/30x6	1
4090/05/07	6302RSH2	Refer to Table 21	1	6302Z	S		20/35x7	1
4100/05	6302			6302Z	S		20/35x7	1
401H	6302			6302Z	S		20/35x7	1
4100/15/25	6304			6305Z	D		32/52x8	1
4130/35	6305			6306	D		38/58x11	1
4145/55	6305R			6306	D		38/58x11	1
4160/65	6307R			6308	D		55/78x12	1
416H	6307R			6308	D		55/78x12	1
4170/75	6406	60UZS-417T2-SX	2	6407	D		60/82x12	1

Table 20 Double Reduction Units Intermediate Shaft Parts

Frame Size	BEARINGS			
	Part No. 5-02	Part No. 5-04	Part No. 5-03	
	Number	Number	Qty	Number
4105DA	6302	Refer to Table 21	1	6007
4115DB	6304			6205
4135DC	6305			6206
4145DB	6305			6206
4165DC	6307R			6208
4175DC	6406	60UZS417T2-SX	2	6208

Table 21 Eccentric Bearing Identification Number

Frame Size	4090 4095 4097	4100 4105 410H 4105DA	4110 4115 4125 4115DB	4130 4135 4135DC	4145 4145DB	4155	4160 4165 4165DC	416H
Ratio/Part No.	3-04							
6	15UZE20906T2	15UZ21006T2	22UZ2110608T2	25UZ21406-11T2	-	25UZ28506-11T2S	35UZ2160608T2S	35UZ2160608T2S
8	15UZE20908-15T2	15UZ21008-15T2	22UZ2110608T2	25UZ21406-11T2	-	25UZ28506-11T2S	35UZ2160608T2S	35UZ2160608T2S
11	15UZE20908-15T2	15UZ21008-15T2	22UZ2111115T2	25UZ21406-11T2	25UZ21406-11T2	25UZ28506-11T2S	35UZ21611-15T2S	35UZ21611-15T2S
13	15UZE20908-15T2	15UZ21008-15T2	22UZ2111317T2	25UZ21413-17T2	25UZ21413-17T2	25UZ28513-17T2S	35UZ21611-15T2S	35UZ21611-15T2S
15	15UZE20908-15T2	15UZ21008-15T2	22UZ2111115T2	25UZ21413-17T2	25UZ21413-17T2	25UZ28513-17T2S	35UZ21611-15T2S	35UZ21611-15T2S
17	15UZE20917T2	15UZ21017T2	22UZ2111317T2	25UZ21413-17T2	25UZ21413-17T2	25UZ28513-17T2S	35UZ21617-25T2S	35UZ21617-25T2S
21	15UZE20921T2	15UZ21021T2	22UZ21121T2	25UZ2142125/417T2	25UZ2142125/417T2	25UZ2852125/417T2S	35UZ21617-25T2S	35UZ21617-25T2S
25	15UZE20925/29T2	15UZ21025/29T2	22UZ21125/29T2	25UZ2142125/417T2	25UZ2142125/417T2	25UZ2852125/417T2S	35UZ21617-25T2S	35UZ21617-25T2S
29	15UZE20925/29T2	15UZ21025/29T2	22UZ21125/29T2	25UZ2142935T2	25UZ2142935T2	25UZ2852935T2S	35UZ2162935T2S	35UZ2162935T2S
35	15UZE20935T2	15UZ21035T2	22UZ21135T2	25UZ2142935T2	25UZ2142935T2	25UZ2852935T2S	35UZ2162935T2S	35UZ2162935T2S
43	15UZE20943T2	15UZ21043T2	22UZ21143T2	25UZ21443-59T2	25UZ21443-59T2	25UZ28543-59T2S	35UZ2164351T2S	35UZ2164351T2S
51	15UZE20951/ 814359T2	15UZ21051/ 824359T2	22UZ2115159T2	25UZ21443-59T2	25UZ21443-59T2	25UZ28543-59T2S	35UZ2164351T2S	35UZ2164351T2S
59	15UZE20959T2	15UZ21059T2	22UZ2115159T2	25UZ21443-59T2	25UZ21443-59T2	25UZ28543-59T2S	35UZ21659T2S	35UZ21659T2S
71	15UZE20971/ 8187T2	15UZ21071/ 8287T2	22UZ2117187T2	25UZ2147187T2	25UZ2147187T2	25UZ2857187T2S	35UZ21671/ 659T2S	35UZ21671/ 659T2S
87	15UZE20987T2	15UZ21087T2	22UZ2117187T2	25UZ2147187T2	25UZ2147187T2	25UZ2857187T2S	35UZ21687T2S	35UZ21687T2S
119	15UZE209119T2	15UZ210119T2	-	-	-	-	-	-

CYCLO – DISASSEMBLY/ASSEMBLY

Disassembly

SM-CYCLO® Reducers are designed to provide maximum ease in disassembling and reassembling ...they require no special maintenance skills.

The following procedures and precautions are recommended at time of disassembly and assembly:

- Work should be performed in a dust-free, humidity-free area.
- Use of a soft or plastic hammer is recommended.
- Care should be taken not to damage parts, i.e. coil, bearings, seals, etc.
- Inspect all components and replace as necessary.
- Be extremely careful in handling of bearings.

1. Remove the complete SM-CYCLO® Reducer with adaptor (motorized type) from the driven machine.
2. Remove the cooling fan cover and fan from those Speed Reducers (not motorized) equipped with a cooling fan, and stand the unit on a solid base with its high speed shaft side down. Remove the through bolts for the high speed end shield, ring gear housing, thus separating the Cyclo from the bevel gear case so that the inner mechanism can be removed (Figs. 11-16).

Note: If the reducer is motorized (C-adaptor and coupling) remove the motor and coupling before following the procedure outlined above. As a final step, remove the adaptor and cooling fan.

3. If the unit will not separate easily, gently drive a wedge at the line X . . . X shown in Fig. 1 on page 8 (if in so doing a burr is produced, be sure to remove it before reassembly).

4. Take out the slow speed shaft rollers, item 1-06, page 8 (Fig. 12). Check the slow speed shaft pins (1-01) to see whether any rollers have adhered to them.

5. The top cycloid disc (2-04) on the slow speed side can be easily lifted out with both hands (Fig. 13).

6. Remove the spacer ring (2-05).

7. The eccentric (3-04) can be removed from the high speed shaft (3-01) after taking out the retaining ring (3-10) and the inner bearing raceway (Fig. 14, 15).

Note: In certain sizes, the eccentric bearings are roller bearings without a retainer. Remove bearings of the top disc before proceeding with the next step.

8. Take out the second disc located on the motor side. (Also remove second disc bearings and eccentric with inner bearing raceway if required.)

9. Remove the ring gear housing (2-01).

10. The high speed shaft (3-01) with bearings is removed from the high speed shaft end shield (8) by tapping the shaft end after first taking off the retaining ring (3-11).

11. The cycloid disc is made from bearing steel and heat treated while the spacer ring is cast iron. Take care not to strike them together while handling.

The above instructions cover complete disassembly. In ordinary cases, however, only the removal of the cycloid discs and the eccentric is necessary.

***Note:** Retaining ring is part of bearing A (Part No. 1-02).

Assembly

SM-CYCLO® Reducers are reassembled by reversing the disassembly procedure. Care must be taken to exclude dust or foreign matter from the moving parts, and to see that gaskets are properly placed to make the assembly oil-tight.

Following are some helpful points to remember when assembling SM-CYCLO® Reducers.

1. Set the ring gear housing and insert the ring gear pins and rollers; then test-rotate the pins and rollers by hand. (Apply grease liberally to the ring gear pins and rollers before they are inserted in grease lubricated SM-CYCLO® Reducers.)
2. Cycloid discs are a matched pair each carrying the same number which is stamped on one side of each disc.
3. Set the cycloid disc with the stamped number face up as shown in Fig. 16.

4. Insert the spacer (3-07) and then insert the eccentric with bearings by rapping with a wooden or hard rubber mallet (Fig. 15).

5. Insert the other spacer and the inner bearing raceway. Secure them with the retaining ring (Fig. 14).

6. Set the spacer ring in place.

7. Insert top disc in such a way that the mark is 180° opposed to the marking of the bottom disc (Fig. 13).

8. Insert slow speed shaft rollers (Fig. 12).

9. Put the slow speed shaft pins into the rollers (Fig. 11). The above instructions are for eccentric bearings with retainer. Following are the instructions suggested for roller bearings without retainer.

- a. First insert the eccentric with inner raceways of bearings by rapping with a wooden or hard rubber mallet.

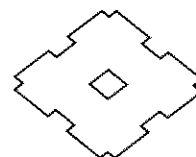


Fig. 11

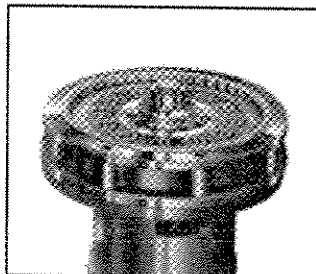


Fig. 12

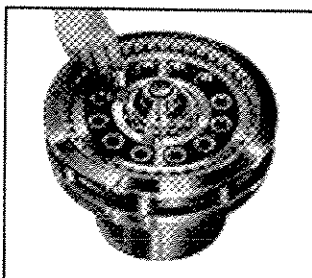


Fig. 13

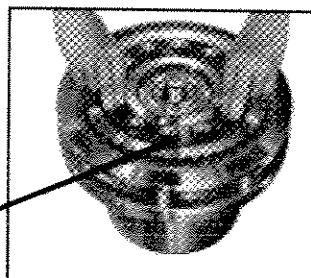


Fig. 14

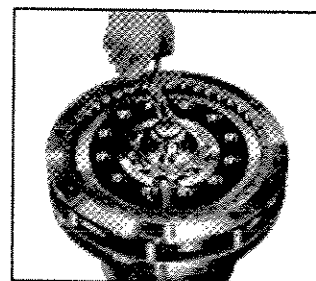


Fig. 15

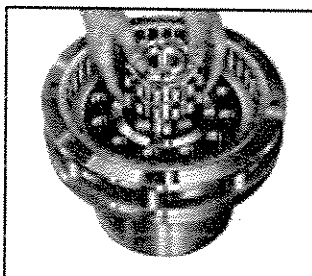
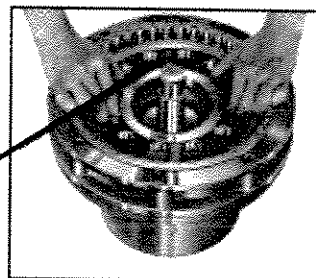


Fig. 16



Note: Insert second disc with number facing slow speed side, exactly 180° opposed to number on first disc.

Note: Set disc with number facing slow speed side.

b. Apply grease to the raceway of the eccentric on the disc. Fix the rollers and set disc in place.

c. Insert the spacer ring and set second disc in such a way that mark is 180° opposed to mark of bottom disc.

Eccentric CAM Replacement Precautions

The eccentric bearings are specially designed for installation on SM-CYCLO® Reducers. They are special roller bearings without outer raceways (refer to the list of bearings on page 11).

It is necessary to insert replacement bearings with numbered surfaces of the inner raceways facing outward. Note that the wrong insertion of the bearings (i.e., insertion of bearings with numbered surfaces inside) causes damage.

Disassembly and Assembly of Sizes 4075-4097 SM-CYCLO® Reducers

Small sizes 4075-4097 are of a single disc system, so they differ in construction from larger sizes in the following ways:

1. A balance weight is provided in lieu of the two-disc system.
2. The balance weight must be positioned exactly 180° as opposed to that of the eccentric.
3. There are no end plates on either side of the eccentric. In all other respects, 4075-4097 have exactly the same construction as the larger sizes. Follow the instructions given under "Disassembly and Assembly".

GENERAL CONSTRUCTION – PARTS LIST

Cyclo Gearmotor Input Section

Fig. 1 Single Reduction

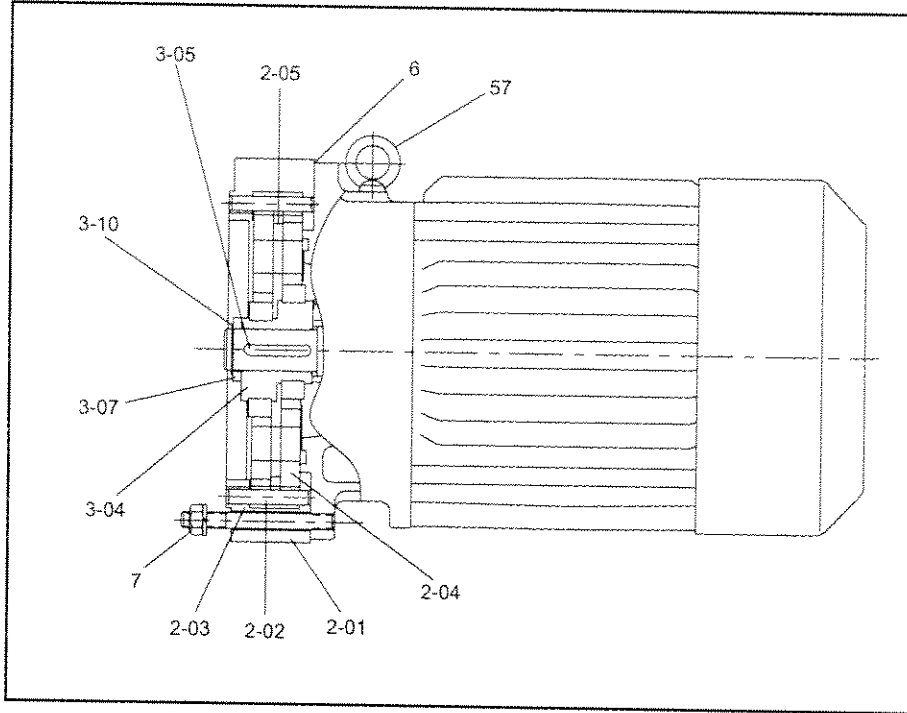


Table 1 Main Parts

Part No.	Part Name
2-01	Ring Gear Housing
2-02	Ring Gear Pins
2-03	Ring Gear Rollers
2-04	Cyclo Disc
2-05	Spacer Ring
3-04	Eccentric Bearing Assembly
3-05	Eccentric Key
3-06	Balance Weight
3-07	Spacer
3-10	Retaining Ring
6	Gasket Set
7	Casing Nuts & Bolts
57	Eye Bolt

BEARINGS, OIL SEALS, GASKETS

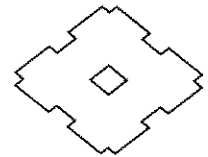


Fig. 9

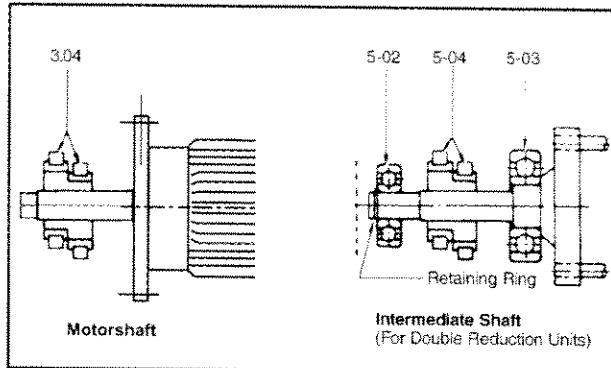


Fig. 10

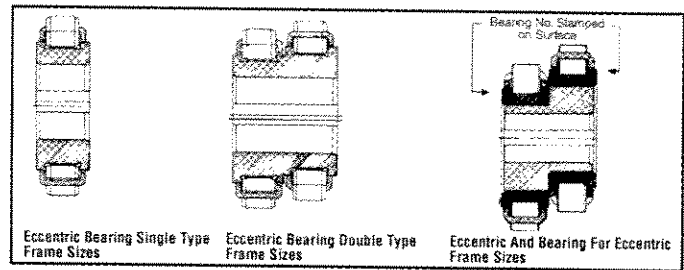


Table 19 Single Reduction Units

Frame Size	BEARING	
	MOTOR SHAFT	
	Part No. 3-04	
	Number	Qty
4085	19UZS208T2	1
4090/05/07	Refer to Table 21	1
4100/05		
410H		
4110/15/25		
4130/35		
4145/55		
4160/65		
416H		
4170/75	60UZS-417T2-SX	2

Table 20 Double Reduction Units Intermediate Shaft Parts

Frame Size	BEARINGS			
	Part No. 5-02	Part No. 5-04	Part No. 5-03	
	Number	Number	Qty	Number
4105DA	6302	Refer to Table 21	1	6007
4115DB	6304			6205
4135DC	6305			6206
4145DB	6305			6206
4165DC	6307R			6208
4175DC	6406	60UZS417T2-SX	2	6208

Table 21 Eccentric CAM Identification Number

Frame Size	4090 4095 4097	4100 4105 410H 4105DA	4110 4115 4125 4115DB	4130 4135 4135DC	4145 4145DB	4155	4160 4165 4165DC	416H
Ratio / Part No.	3-04							
6	15UZE20906T2	15UZ21006T2	22UZ2110608T2	25UZ21406-11T2	--	25UZ8506-11T2S	35UZ2160608T2S	35UZ2160608T2S
8	15UZE20908-15T2	15UZ21008-15T2	22UZ2110608T2	25UZ21406-11T2	--	25UZ8506-11T2S	35UZ2160608T2S	35UZ2160608T2S
11	15UZE20908-15T2	15UZ21008-15T2	22UZ2111115T2	25UZ21406-11T2	25UZ21406-11T2	25UZ8506-11T2S	35UZ21611-15T2S	35UZ21611-15T2S
13	15UZE20908-16T2	15UZ21008-16T2	22UZ2111317T2	25UZ21413-17T2	25UZ21413-17T2	25UZ8513-17T2S	35UZ21611-15T2S	35UZ21611-15T2S
15	15UZE20908-16T2	15UZ21008-16T2	22UZ2111115T2	25UZ21413-17T2	25UZ21413-17T2	25UZ8513-17T2S	35UZ21611-15T2S	35UZ21611-15T2S
17	15UZE20917T2	15UZ21017T2	22UZ2111317T2	25UZ21413-17T2	25UZ21413-17T2	25UZ8513-17T2S	35UZ21617-25T2S	35UZ21617-25T2S
21	15UZE20921T2	15UZ21021T2	22UZ21121T2	25UZ2142125/417T2	25UZ2142125/417T2	25UZ852125/417T2S	35UZ21617-25T2S	35UZ21617-25T2S
25	15UZE2092529T2	15UZ2102529T2	22UZ2112529T2	25UZ2142125/417T2	25UZ2142125/417T2	25UZ852125/417T2S	35UZ21617-25T2S	35UZ21617-25T2S
29	15UZE2092529T2	15UZ2102529T2	22UZ2112529T2	25UZ2142935T2	25UZ2142935T2	25UZ852935T2S	35UZ2162935T2S	35UZ2162935T2S
35	15UZE20935T2	15UZ21035T2	22UZ21135T2	25UZ2142935T2	25UZ2142935T2	25UZ852935T2S	35UZ2162935T2S	35UZ2162935T2S
43	15UZE20943T2	15UZ21043T2	22UZ21143T2	25UZ21443-59T2	25UZ21443-59T2	25UZ8543-59T2S	35UZ2164351T2S	35UZ2164351T2S
51	15UZE20951/ 814359T2	15UZ21051/ 824359T2	22UZ2115159T2	25UZ21443-59T2	25UZ21443-59T2	25UZ8543-59T2S	35UZ2164351T2S	35UZ2164351T2S
59	15UZE20959T2	15UZ21059T2	22UZ2115159T2	25UZ21443-59T2	25UZ21443-59T2	25UZ8543-59T2S	35UZ21659T2S	35UZ21659T2S
71	15UZE20971/ 8187T2	15UZ21071/ 8287T2	22UZ2117187T2	25UZ2147187T2	25UZ2147187T2	25UZ857187T2S	35UZ21671/ 659T2S	35UZ21671/ 659T2S
87	15UZE20987T2	15UZ21087T2	22UZ2117187T2	25UZ2147187T2	25UZ2147187T2	25UZ857187T2S	35UZ21687T2S	35UZ21687T2S
119	15UZE209119T2	15UZ210119T2	--	--	--	--	--	--

DISASSEMBLY/ASSEMBLY OF GEARMOTOR

Disassembly

SM-CYCLO® Gearmotors are designed to provide maximum ease in disassembly and reassembly... they require no special maintenance skills.

The following procedures and precautions are recommended at time of disassembly and assembly:

- Work should be performed in a dust-free, humidity-free area.
 - Use of a soft or plastic hammer is recommended.
 - Care should be taken not to damage parts, i.e. coil, bearings, seals, etc.
 - Inspect all components and replace as necessary.
 - Be extremely careful in handling of bearings.
1. Remove the complete SM-CYCLO® Gearmotor from the driven machine.
 2. Place the gearmotor vertically with the output shaft upward.
 3. Remove the through bolts from the motor flange, ring gear housing, and lift the slow speed side, thus separating the unit into two parts so that the inner mechanism can be removed (Fig. 18-23).
 4. If the unit will not separate easily, gently drive a wedge at the line X . . . X shown in Fig. 1 on page 12 (if in so doing a burr is produced, be sure to remove it before reassembly).
 5. Take out the slow speed shaft rollers, item 1-06, page 12 (Fig. 19). Check the slow speed shaft pins (1-01) to see whether any rollers have adhered to them.

6. The top cycloid disc (2-04) on the slow speed side can be easily lifted out with both hands (Fig. 20).

7. Remove the spacer ring (2-05).

8. The eccentric bearing assembly (3-04) can be removed from the motor shaft after taking out the retaining ring (3-10), or the bearings (3-02), figures 21, 22.

Note, in certain sizes, the eccentric bearings are roller bearings without a retainer. Remove rollers of the top disc and the second disc on the motor side before removing the eccentric.

9. Take out the second disc located on the motor side.

10. Remove the ring gear housing (2-01) from the motor.

11. The cycloid disc is made from bearing steel and is heat treated, while the spacer ring is cast iron. Take care not to strike them together while handling.

***Note:** If motor is in need of repair, please send to any authorized EASA shop.

Assembly

SM-CYCLO® Gearmotors are reassembled by reversing the disassembly procedure. Care must be taken to exclude dust or foreign matter from the moving parts, and to see that gaskets are properly placed to make the assembly oil-tight.

Following are some helpful points to remember when assembling SM-CYCLO® Gearmotors.

1. Set the ring gear housing and insert the ring gear pins and rollers; then test-rotate the pins and rollers by hand. (Apply grease liberally to the ring gear pins and rollers before they are inserted in grease lubricated SM-CYCLO® Gearmotors).
2. Cycloid discs are a matched pair each carrying the same number which is stamped on one side of each disc.
3. Set the cycloid disc with the stamped number face up as shown in figure 23.

4. Insert the eccentric bearing by rapping with a wooden or hard rubber mallet (Fig. 22).

5. Insert the spacer and the inner bearing raceway. Secure them with the retaining ring (Fig. 21).

6. Set the spacer ring in place.

7. **Insert top disc in such a way that the mark is 180° opposed to the marking of the bottom disc (Fig. 20).**

8. Insert slow speed shaft rollers (Fig. 19).

9. Put the slow speed shaft pins into the rollers (Fig. 18). The above instructions are for eccentric bearings with retainer. Following are the instructions suggested for roller bearings without retainer:

a. First insert the eccentric with inner raceways of bearings by rapping with a wooden or hard rubber mallet.

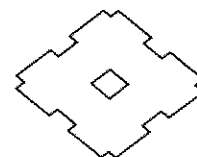


Fig. 18

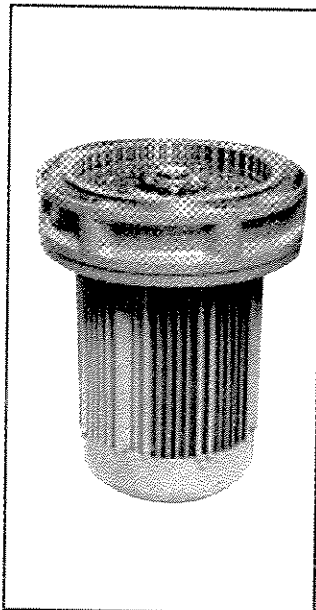


Fig. 19

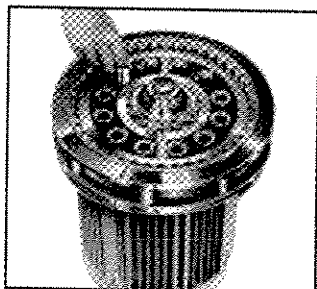


Fig. 22

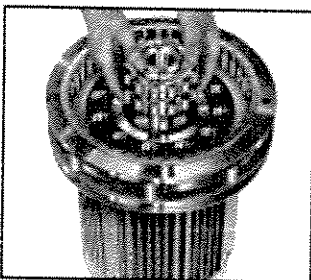


Fig. 20

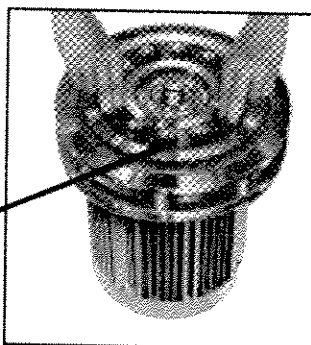


Fig. 21

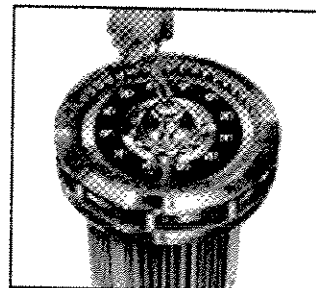
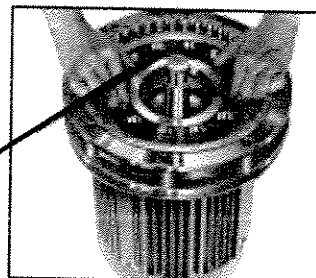


Fig. 23



Note: Insert second disc with number facing slow speed side, exactly 180° opposed to number on first disc.

Note: Set disc with number facing slow speed side.

b. Apply grease to the raceway of the eccentric on the disc. Fix the rollers and set disc in place.

c. Insert the spacer ring and set second disc in such a way that mark is 180° opposed to mark of bottom disc.

Eccentric Bearing Replacement Precautions

The eccentric bearings are specially designed for installation on SM-CYCLO® Reducers. They are special roller bearings without outer raceways (refer to the list of bearings on page 9).

It is necessary to insert replacement bearings with numbered surfaces of the inner raceways facing outward. Note that the wrong insertion of the bearings (i.e., insertion of bearings with numbered surfaces inside) causes damage.

Disassembly and Assembly of Sizes 4085-4097 SM-CYCLO® Reducers

Small sizes 4085-4095 are of a single disc system, so they differ in construction from larger sizes in the following ways:

1. A balance weight is provided in lieu of the two-disc system.
2. The balance weight must be positioned exactly 180° as opposed to that of the eccentric.
3. There are no end plates on either side of the eccentric. In all other respects, they have exactly the same construction as the larger sizes. Follow the instructions given under "Disassembly and Assembly".

STANDARD INTEGRAL MOTOR CHARACTERISTICS

Motor Characteristics

The SM-Bevel Buddybox gearmotors' full load ratings and amperage can be found below in table M-1. These ratings are based on the motor's design values. If additional information is required, please consult factory.

Table M-1 — 230/460 Volt, Synchronous Speed 1800 rpm, 60 Hz, Continuous Duty, TEFC

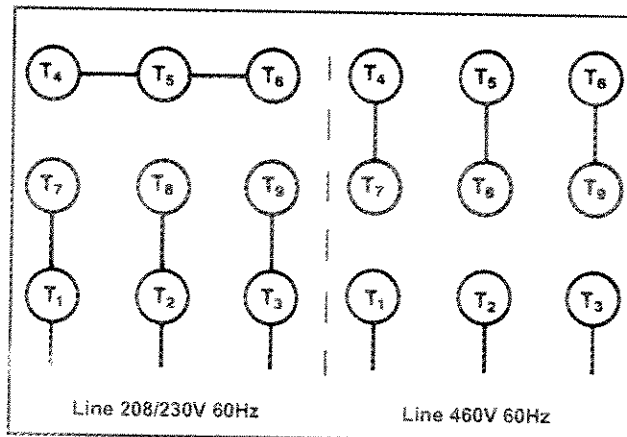
HP	Frame Size	Full Load rpm	Torque in. lb.	Full Load Current Amperage			Starting Current Amperage		Torque % of F.L. Break-down		Efficiency %	Power Factor %	Code Letter*	Inertia WR ² lb. ft ²
				230V	460V	208V	230V	460V	Starting	Break-down				
1/2	F-71M	1740	18.3	2.1	1.1	2.0	9.8	4.9	295	280	71.9	65.1	J	0.0154
3/4	F-80S	1730	27.4	2.5	1.2	2.5	12.3	6.2	266	245	76.9	73.0	H	0.0227
1	F-80	1750	36.0	3.4	1.7	3.4	17.6	8.8	269	303	77.2	72.3	H	0.0285
1.5	F-90S	1730	54.7	4.6	2.3	4.7	28.6	14.3	273	281	80.3	74.1	J	0.0451
2	F-90L	1740	72.5	6.0	3.0	6.2	36.8	18.4	263	270	82.3	75.6	J	0.0504
3	F-100L	1730	109	8.4	4.2	8.7	54.8	27.4	277	266	84.4	77.2	J	0.0789
5	F-112M	1730	183	13.0	6.5	13.7	91.5	45.8	308	279	86.3	82.4	J	0.201
7.5	F-132S	1710	277	18.1	9.0	20.1	120	61	223	221	86.7	88.2	H	0.271
10	F-132M	1740	361	23.6	11.8	26.4	147	73.5	212	214	89.6	88.9	G	0.635
15	F-160M	1740	542	34.3	17.2	38.2	231	115	248	221	90.5	89.0	G	0.891
20	G-160L	1740	725	45.8	22.9	51	272	136	222	220	91.6	89.9	F	2.13
25	G-180M	1770	891	57	28.4	63	343	171	199	235	92.6	88.2	F	5.34
30	G-180M	1760	1075	68	34.2	77	388	194	192	226	91.5	88.1	F	5.34

*Code letter shown is for 230V or 460V operation. Consult factory for other voltages.

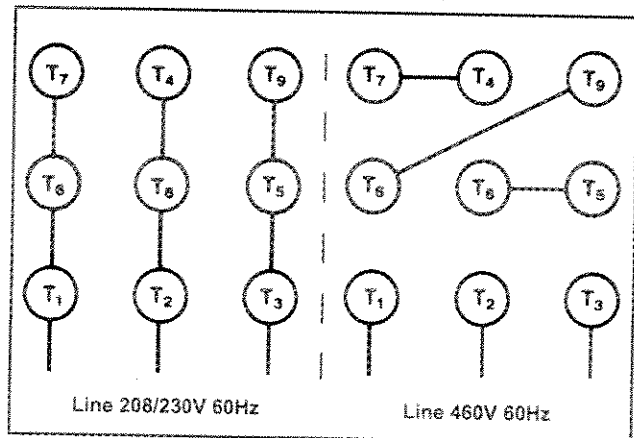
Standard Wiring Diagram — 208, 230/460V

Illustrated below are the wiring diagrams for our standard motor. For additional information please refer to motor name plate. Due to changes in design features, this diagram may not always agree with that on the motor. If different, the motor diagram found inside the conduit box cover is correct.

Y-Connected (5 HP and smaller)



Delta-Connected (7½ HP and larger)



GENERAL CONSTRUCTION OF MOTOR

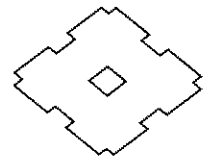


Fig. 11 Motor construction

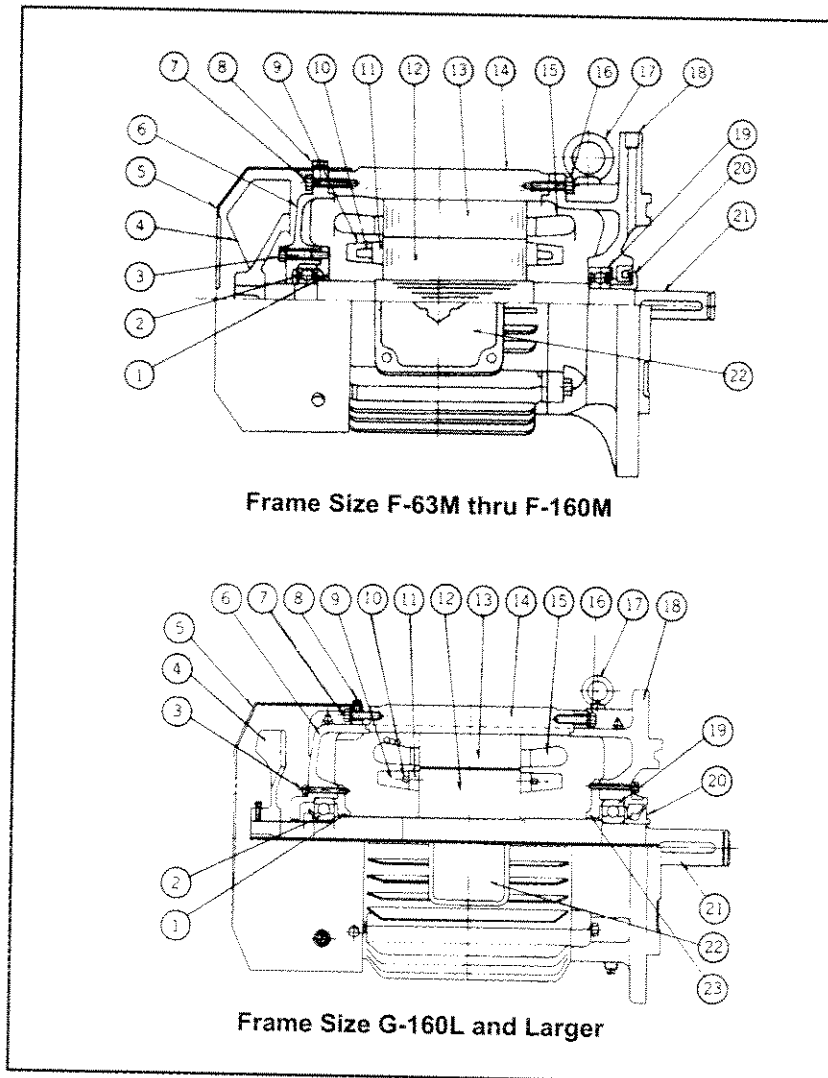


Table 22 Main Parts

Parts No.	Parts Name
1	Bearing Cover**
2	Bearing
3	Bolt**
4	Fan*
5	Fan Cover
6	End Bracket
7	Bolt
8	Bolt
9	Internal Fan
10	Hub
11	Short Circuit Ring
12	Rotor Core
13	Stationary Core
14	Stator Frame
15	Stator Winding
16	Bolt
17	Eye Bolt
18	Cyclo Flange Bracket
19	Bearing
20	Slinger/Oil Seal
21	Motor Shaft
22	Conduit Box
23	Bearing Cover
24	Bearing Sleeve

* No Fan 1/8 H.P., F-63S frame.
 ** F-132M and larger.

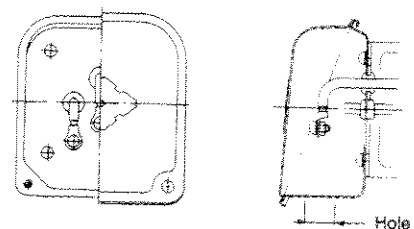


Fig. 12 Conduit Box-Standard Type

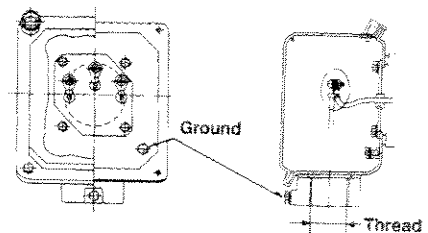


Fig. 13 Weather-Proof Type

GENERAL INSPECTION OF MOTOR

The following items should be verified when the motor is received:

1. Check the nameplate horsepower rating (H.P.), number of poles (P), type, voltage (volt) and frequency (Hz).

2. Rotate the motor shaft by hand to check for binding.

3. Check the motor's overall appearance for possible shipping damage.

Pre-Start Up Inspection

Check the following prior to start-up:

1. **Wiring:** Prior to wiring refer to the name plate affixed to the motor portion of the gearmotor. Check power supply, interconnects, relays protective starting devices, [i.e. Star (Wye) delta — if reduced voltage starting is required], space heaters, thermal sensors and other accessories.

2. **Grounding: Caution:** The motor frame and conduit box must be properly grounded so as to avoid electrical shock.

3. **Insulation Resistance:** Stator winding measurements are to be made at the motor terminals. Stator windings of less than 600 volts are to be measured with a 500 volts megger.

Insulation resistance will vary depending on winding temperature, moisture, cleanliness, duration of usage, and test voltage and charging period.

Other factors which will affect insulation resistance are: output rating, voltage, insulation class and RPM. Although it is difficult to decide a minimum value applicable to each and every case, for simplicity the following may be used as a guide:

1 Mega ohms for less than 600 volts

If the measured insulation resistance is less than indicated above, the cause may be due to moisture. The resistance can therefore be increased by subjecting wire to heat, hot air, vacuum, current (short circuit current, low voltage — no load current, or direct current).

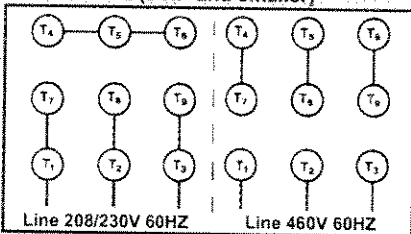
If the Mega ohm reading cannot be recovered by drying — consult factory.

Fig. 14 Typical Wiring Diagrams

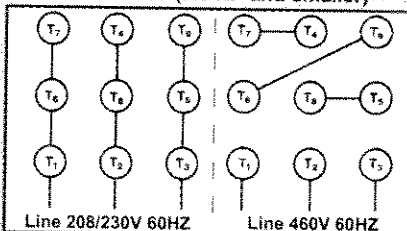
Illustrated below are the wiring diagrams for our standard motor, for additional information please refer to motor name plate. If diagram in conduit box conflicts with diagrams listed below, please refer to diagram in motor conduit box.

THREE PHASE MOTORS

Y-Connected (5 HP and smaller)

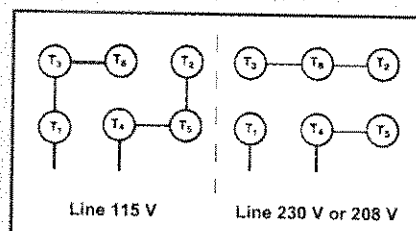


Delta-Connected (7.5 HP and smaller)



To reverse rotation interchange any two line leads.

SINGLE PHASE MOTORS



To reverse rotation interchange leads T5 and T8.

Preparation and Start-up

Prior to start-up, please check the following under no load:

1. The driven load and the SM-Cyclo gearmotor are properly secured prior to operation.
2. The motor bearings are grease packed when shipped from our factory, however, if the motor is operated after long term storage, you must replenish the grease in the open type bearing only. Please refer to page 19, Table 24 for correct quantity.
3. Check the direction of rotation. If a reverse direction is

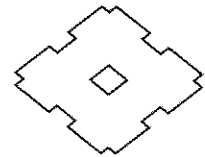
required, simply reverse any two power leads.

4. Check the voltage supply and current (line and phase) to verify balancing for a 3 phase power source.

5. When power is supplied to the motor and the starting is abnormally long, starting is not completed, or any abnormal sound is heard — immediately shut off the power and consult factory.

6. Measure the current draw. The current measured at full load should not exceed the nameplate rating.

BEARINGS, OIL SEALS FOR MOTOR



Non-Contact Shield Type Ball Bearings

These bearings do not require grease replenishment, however, it is recommended that they be replaced once every three years when operated under normal conditions, and once a year under severe duty conditions.

Open Type Ball or Roller Bearing

For interval of grease replenishment, quantities of grease and the type of grease for the open type bearing refer to tables 23 and 24.

Table 23 Type of Grease for Open Type Bearings

Class of Insulation	Grease
B	Shell Alvania No. 2
H, F	Shell Darina No. 2

**Table 24
Interval and Quantity of Grease Replenishment**

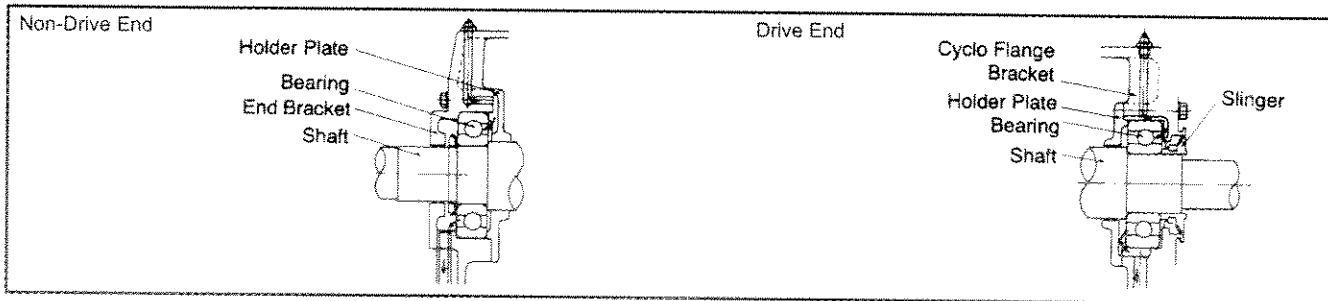
Bearing Size	Replenish-ment Qty (oz)	Initial Qty (oz)	Interval of Replenishment (HR)		
			900 rpm	1200 rpm	1800 rpm
6314	1.4	7.1	7000	5000	2500
6315	1.6	8.1	6500	4500	2500
6316	1.8	9.2	6500	4500	2500
6317	1.9	10	6000	4000	2000
6318	2.1	12	5500	4000	2000
6319	2.3	14	5500	3500	1500
6320	2.5	16	5000	3500	1500
6321	2.6	18	5000	3000	1500
6322	2.8	19	4500	3000	1000
6324	3.5	25	4000	2500	1000
6412	1.4	7.1	7000	5000	3000
6413	1.6	8.1	6500	4500	2500
6414	1.9	10	6500	4500	2500
NU314	1.4	4.2	3500	2500	—
NU315	1.6	5.3	3000	2000	—
NU316	1.8	7.1	3000	2000	—
NU317	1.9	8.8	3000	2000	—
NU318	2.1	10	2500	2000	—
NU319	2.3	12	2500	1500	—
NU320	2.5	14	2500	1500	—
NU321	2.6	16	2500	1500	—
NU322	2.8	18	2000	1500	—
NU324	3.5	23	2000	1000	—

Grease Replenishment Procedure

The gearmotor must be running with the drain plug removed. Using a grease gun, inject a quantity of grease until some grease is purged from the drain. Plug the drain approximately 10 minutes after start of operation.

Note: Excess grease replenishment may cause overheating or leakage. A lack of grease will cause premature failure.

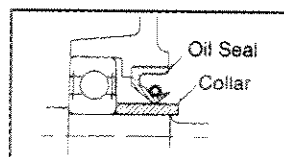
Fig. 15 Typical Bearing Assembly



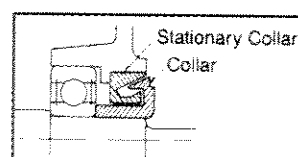
Oil Seal

The reducer and the motor of the SM-CYCLO® Gear motor are integrally assembled. In grease lubricated units, a single lip seal is installed between the motor section and reducer section as shown in figure 16. On oil lubricated units a slinger type oil seal is used — figure 17.

**Fig. 16
Single Lip Type:
Grease Lubed Units**



**Fig. 17
Slinger Type:
Oil Lubed Units**



Note: For oil seal replacement sizes see table 25.

BEARINGS AND OIL SEALS

Table 25 Bearings and Oil Seals Sizes†

MOTOR FRAME NO.	MOTOR		CYCLO FRAME SIZE	BEARING		OIL SLINGER PART NO.	OIL SEAL SIZE**
	HP	RPM		NON-DRIVE END	DRIVE END		
F-63S	¼	1800	4085	6202ZZ-CM	6302ZZ-CM	—	S20358
			4090/95/97	6202ZZ-CM	6303ZZ-CM	—	S22357
F-63M	¼ ⅓	1800	4085	6202ZZ-CM	6302ZZ-CM	—	S20358
			4090/95/97	6202ZZ-CM	6303ZZ-CM	—	S22357
			4100/05	6202ZZ-CM	6303ZZ-CM	—	S22357
F-71M	½	1800	4085	6202ZZ-CM	6302ZZ-CM	—	S20358
			4090/95/97	6202ZZ-CM	6303ZZ-CM	—	S22357
			4100/05	6202ZZ-CM	6303ZZ-CM	—	S22357
			4110/15	6202ZZ-CM	6205ZZ-CM	—	S30458
F-80S F-80M	¾ 1	1800	4090/95/97	6203ZZ-CM	6303ZZ-CM	—	S22357
			4100/05	6203ZZ-CM	6303ZZ-CM	—	S22357
			4110/15	6203ZZ-CM	6205ZZ-CM	—	S30458
			4130/35	6203ZZ-CM	6206ZZ-CM	CD4-4501-G01	(S406212)
F-90S F-90L	1½ 2	1800	4095/97	6204ZZ-CM	6303ZZ-CM	—	S25458
			4100/05/0H	6204ZZ-CM	6204ZZ-CM	—	S26428
			4110/15/25	6204ZZ-CM	6205ZZ-CM	—	S30458
			4130/35 4145/55	6204ZZ-CM	6206ZZ-CM	CD4-4501-G01	(S406212)
			4160/65	6204ZZ-CM	6307ZZ-CM	CD4-4502-G01	(S456812)
F-100L	3	1800	410H	6205ZZ-CM	6204ZZ-CM	—	S26428
			4110/15/25	6205ZZ-CM	6205ZZ-CM	—	S30458
			4130/35 4145/55	6205ZZ-CM	6206ZZ-CM	CD4-4501-G01	(S406212)
			4160/65/6H	6205ZZ-CM	6307ZZ-CM	CD4-4502-G01	(S456812)
F-112M	5	1800	4110/15/25	6206ZZ-CM	6305ZZ-CM	—	—
			4130/35 4145/55	6206ZZ-CM	6206ZZ-CM	CD4-4501-G01	(S406212)
			4160/65/6H	6206ZZ-CM	6307ZZ-CM	CD4-4502-G01	(S456812)
			4170/75	6206ZZ-CM	6308ZZ-CM	CD4-4503-G01	(S508014)
F-132S	7½	1800	4125	6206ZZ-CM	6206ZZ-CM	—	(S30458)
			4130/35 4145/55	6206ZZ-CM	6206ZZ-CM	CD4-4501-G01	(S406212)
			4160/65/6H	6206ZZ-CM	6307ZZ-CM	CD4-4502-G01	(S456812)
			4170/75	6206ZZ-CM	6308ZZ-CM	CD4-4503-G01	(S508014)
F-132M	10	1800	4130/35 4145/55	6307ZZ-CM	6306ZZ-CM	CD4-4501-G01	(S406212)
			4160/65/6H	6307ZZ-CM	6307ZZ-CM	CD4-4502-G01	(S456812)
			4170/75	6307ZZ-CM	6309ZZ-CM	CD4-4504-G01	(S558514)
F-160M	15	1800	4155	6307ZZ-CM	6306ZZ-CM	CD4-4501-G01	(S406212)
			4160/65/6H	6307ZZ-CM	6307ZZ-CM	CD4-4502-G01	(S456812)
			4170/75	6307ZZ-CM	6309ZZ-CM	CD4-4504-G01	(S558514)
G-160L	20 15	1800 1200	4155	6309ZZ-CM	6306ZZ-CM	CD4-4501-G01	(S406212)
			4160/65/6H	6309ZZ-CM	6308ZZ-CM	CD4-4502-G01	(S456812)
			4170/75	6309ZZ-CM	6309ZZ-CM	CD4-4504-G01	(S558514)
F-180MG	25 30 20	1800 1200	4165/6H	6312ZZ-CM	6308ZZ-CM	CD4-4502-G01	(S456812)
			4175	6312ZZ-CM	6309ZZ-CM	CD4-4504-G01	(S558514)

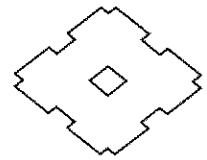
†For 6 pole motors not shown, please consult factory.

Oil slinger is used on oil lubricated cyclo gearmotors.

**Oil seal is used on grease lubricated cyclo gearmotors. Where oil lubrication is standard, oil seal is shown in ().

SM-BEVEL BUDDYBOX

TROUBLE-SHOOTING AND REPAIR



This trouble shooting guide is to help you identify and overcome common problems of reducers and motors. If you have a problem not listed below, please consult factory.

PROBLEM WITH THE REDUCER		POSSIBLE CAUSES	SUGGESTED REMEDY
Runs Hot	Overloading	Load exceeds the capacity of the reducer.	Check rated capacity of reducer, replace with unit of sufficient capacity or reduce load.
	Improper Lubrication	Insufficient lubrication.	Check lubricant level and adjust up to recommended levels.
		Excessive lubrication.	Check lubricant level and adjust down to recommended level.
		Wrong lubricant.	Flush out and refill with correct lubricant as recommended.
Vibration or Noise	Loose Foundation Bolts	Weak mounting structure.	Inspect mounting of reducer. Tighten loose bolts and/or reinforce mounting & structure.
		Loose hold down bolts.	Tighten bolts.
	Worn Disc	Overloading unit may result in damage to disc.	Disassemble and replace disc. Re-check rated capacity of reducer.
	Failure of Bearings	May be due to lack of lubricant.	Replace bearing. Clean and flush reducer and fill with recommended lubricant.
		Overload.	Check rated capacity of reducer, replace with unit of sufficient capacity or reduce load.
	Insufficient Lubricant	Level of lubricant in the reducer not properly maintained.	Check lubricant level and adjust to factory-recommended level.
Output Shaft Does Not Turn	Motor Shaft Broken	Overloading of reducer can cause damage.	Replace broken shaft. Check rated capacity of reducer.
		Key missing or sheared off on input shaft.	Replace key.
	Eccentric Bearing Broken	Lack of lubricant.	Replace eccentric bearing. Flush and refill with recommended lubricant.
	Motor Doesn't Turn	Motor	Refer to page 22.
Oil Leakage	Worn Seals	Caused by dirt or grit entering seal.	Replace seals. Breather filter may be clogged. Replace or clean filter.
	Leakage Into Motor	Overfilled reducer.	Check lubricant level and adjust to recommended level.
		Vent clogged.	Clean or replace element, being sure to prevent any dirt from falling into the reducer.
		Improper mounting position, such as wall or ceiling mount of horizontal reducer.	Mount horizontally or rework reducer to wall or ceiling mount.

SM-BEVEL BUDDYBOX

TROUBLE-SHOOTING AND REPAIR

PROBLEM WITH THE MOTOR		POSSIBLE CAUSES	SUGGESTED REMEDY
Load is disconnected but motor doesn't rotate	Makes a groaning sound	Faulty switch contact	Adjust the contact
		Blown fuse	Replace
		One phase wire of the power supply open	Replace
		Stator coil open	Repair by rewinding or replacing stator assembly
		Stator and rotor touching due to bearing housing wear	Replace the bearing and bracket
	Starts in either direction when turned by hand	Three-phase is operating as single-phase	Check the power source with a voltmeter
	Doesn't make any noise	Stator coil open	Repair by rewinding or replacing stator assembly
		Outside the motor Power failure Open connection wire Faulty switch contact Faulty starter contact	Contact the power company Check the source wiring Adjust the contact
Rotates with the load disconnected, but:	Rotates in the wrong direction	Connection error	Change any two of the three phase source
	Fuse blows	Shorted lead wire	Replace
	Speed doesn't increase	Faulty starter contact	Adjust
	Groans	Overcurrent/Overheating	Rotor and stator touching Repair by rewinding or replacing stator assembly
		Over-current	One phase of stator coil shorted Replace the stator winding
Rotates when the load is disconnected but when the load is connected it:	Switch overheated	Insufficient switch capacity	Replace with one having the rated capacity
		Overload	Drop to the rated load
	Fuse blows	Insufficient fuse capacity	Replace with one having the rated capacity
	Overheats	Overload	Drop to the rated load
		Voltage drop	Consult with the power company
	Speed suddenly drops	Voltage drop	Consult with the power company
		Overload	Drop to the rated load
	Stops	Bearing damaged by overheat	Replace the bearing

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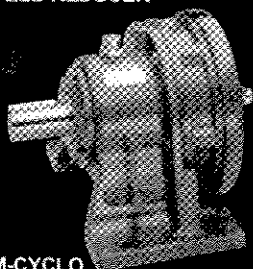
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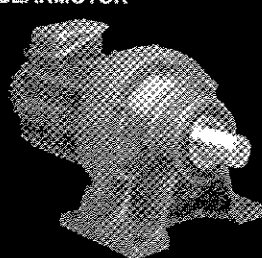
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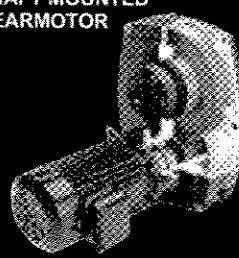


SPEED REDUCER

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GEARMOTOR

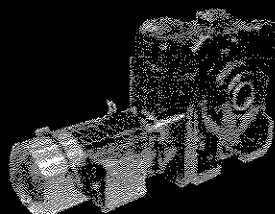
SM-CYCLO
Concentric

**SHAFT MOUNTED
GEARMOTOR**

SM-BUDDYBOX
Parallel Offset

BEVEL GEAR MOTOR

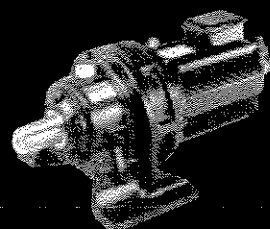
SM-BEVEL BUDDYBOX
Right Angle

WORM GEAR MOTOR

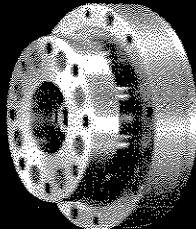
SM-WORM BUDDYBOX
Right Angle

LOW RATIO PLANETARY

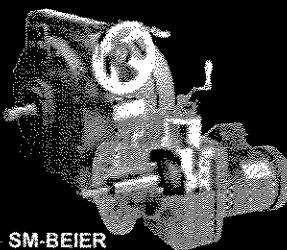
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SERVO GEAR MOTOR

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ZERO BACKLASH CYCLO

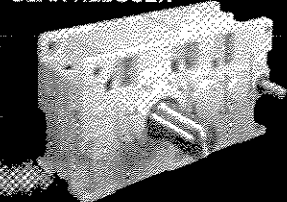
SM-SERVO-MATCH
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**MECHANICAL
VARIABLE SPEED**

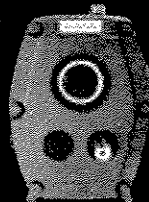
SM-BEIER
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**ELECTRICAL
VARIABLE SPEED**

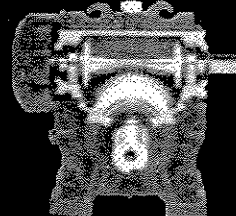
NTAC-2000 **AF-3000**
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**HELICAL
GEAR REDUCER**

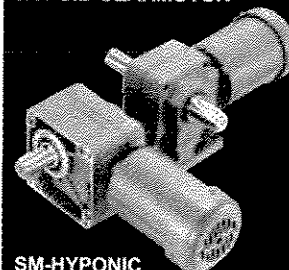
PARAMAX
Parallel Offset &
Right Angle

**SHAFT MOUNT
SPEED REDUCER**

SM-SHAFT MOUNT
Parallel Offset

**DOUBLE ENVELOPING
WORM GEAR**

SM-HEDCON
Right Angle

HYPOID GEAR MOTOR

SM-HYPONIC
Right Angle

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